



Sri Lanka Institute of Information Technology

Assignment 2

Data Warehouse & Business Intelligence

Submitted by:

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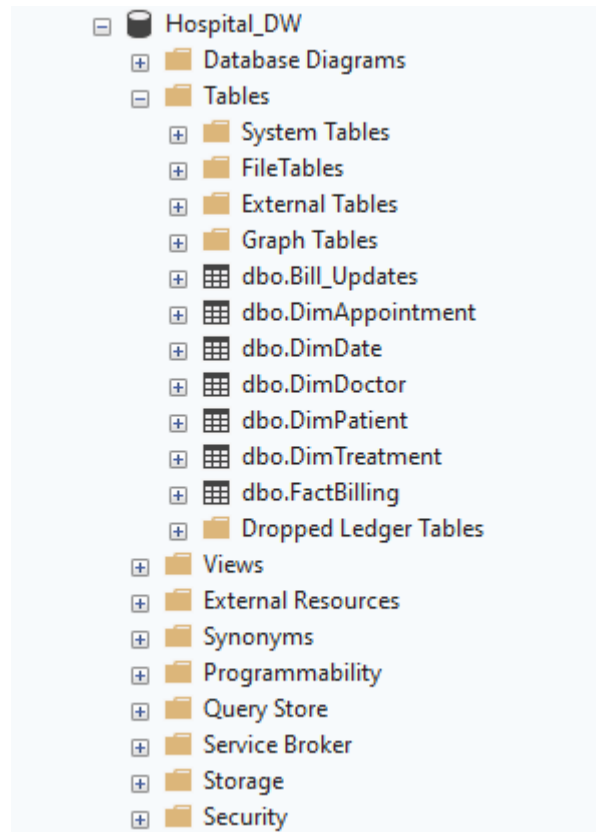
Contents

1) Data source for the Assignment 2	4
2)SSAS Cube implementation	7
2.1 Data Source View.....	8
2.2 Cube Structure	8
2.3 HIERARCHY	11
2.4 KPI Implemented in the Cube.....	13
2) Demonstration of OLAP operations	15
3.1 SLICE Operation	17
Revenue by Hospital Branch (Filtered by Doctor Specialization).....	17
Pivot Chart – Revenue Analysis by Specialization and Location	17
3.2 Dice Operation	18
Revenue by Hospital Branch (Filtered by Specialization and Year).....	18
Revenue Analysis by Specialization, Year, and Location	19
3.3 Pivot Operation	20
Revenue by Hospital Branch and Year (Initial View).....	20
Revenue by Year and Hospital Branch (Pivoted View)	20
3.4 Drill-Down and Roll-Up Operation	21
Drill-Down Operation.....	21
4) PowerBI Reports.....	25
4.1 Report 1 – Revenue Structure Dashboard (Matrix-Based)	25

Supporting Visuals	26
4.2 Report 2 - Operational Performance Dashboard (Interactive Analysis with Multiple Slicers)	29
4.3 Report 3 - Time-Based Analysis Dashboard (Drill-Down).....	35
Dashboard Overview	35
Drill-Down Implementation (Core Feature)	35
4.4 Report 4 - Detailed Analysis Dashboard (Drill-Through)	39
Step 1 – Configuring Drill-Through Field	39
Step 2 – Initiating Drill-Through from Main Dashboard.....	39
Step 3 – Drill-Through Navigation Result.....	40
Step 4 – Back Navigation	41
Step 5 – Visual Design (Detailed Data Presentation)	41

1) Data source for the Assignment 2

The data source for this assignment is the data warehouse created in Assignment 1. It stores hospital-related data such as patients, doctors, appointments, and billing in a structured format. The warehouse uses fact and dimension tables, where the fact table stores measurable data and dimension tables store descriptive details. Data was loaded using an ETL process in SSIS. This data warehouse is used as the main source for analysis in Assignment 2.



Hospital_DW is designed as a star schema with one fact table and five dimension tables, including a Date dimension. The grain of the FactBilling table is defined as one record per billing transaction per patient per treatment.

Dimension Tables – DimPatient

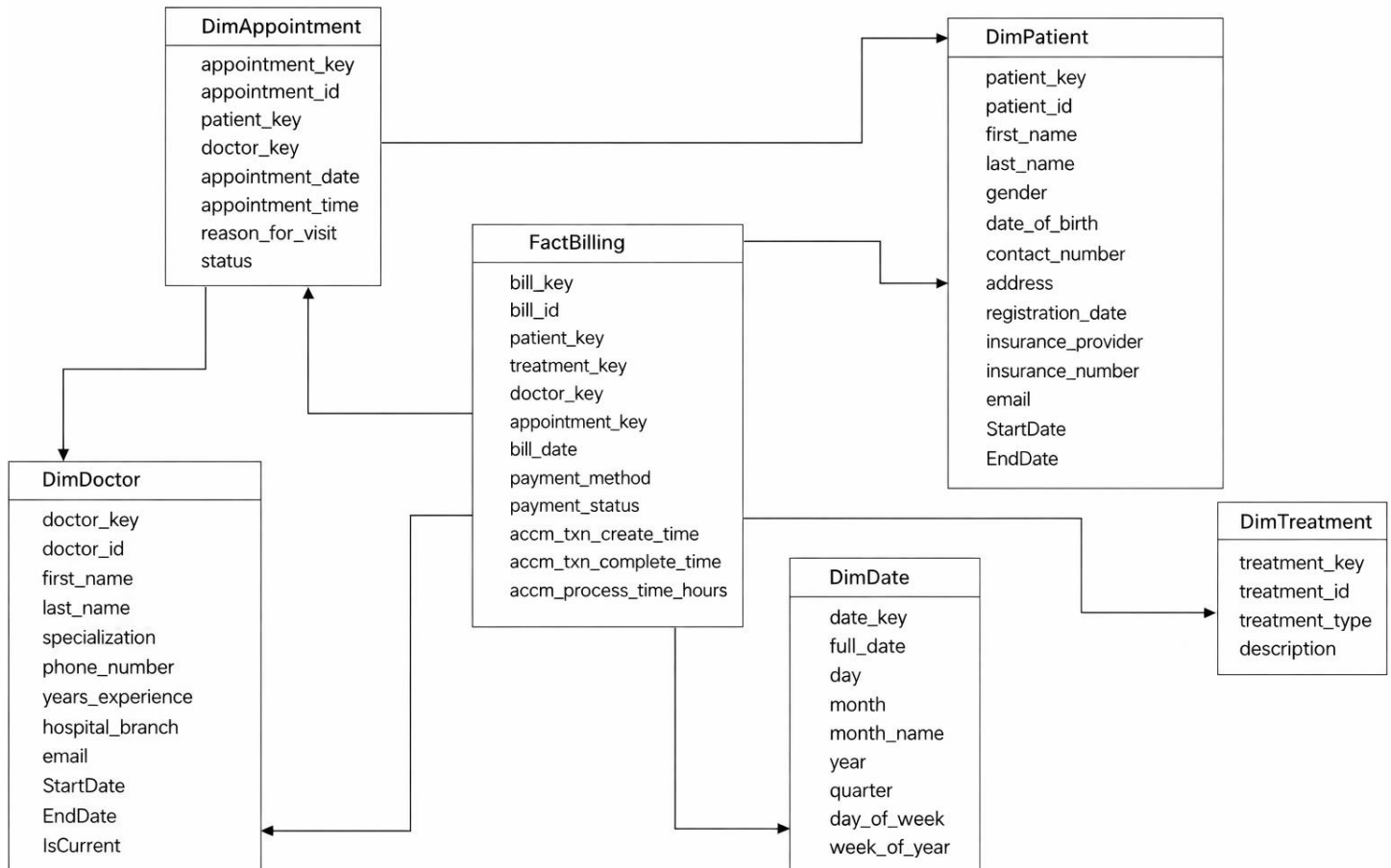
Fact Tables - FactBilling

DimDoctor

DimTreatment

DimAppointment

DimDate



Slowly Changing Dimensions (SCD) are used to manage changes in dimension tables over time. In this project, both DimPatient and DimDoctor are implemented using a combination of:

- SCD Type 1 → Overwrite old values
- SCD Type 2 → Maintain historical records

The SCD logic is implemented using the Slowly Changing Dimension Wizard in SSIS.

Patient Dimension

SCD Type 2 (Keep History) - These values change over time and historical tracking is important.

Applied for – address

insurance_provider

SCD Type 1 (Overwrite) - Only the latest value is required.

Applied for – contact_number

Email

Doctor Dimension

SCD Type 2 (Keep History) - These values change over time and historical tracking is important.

Applied for – specialization

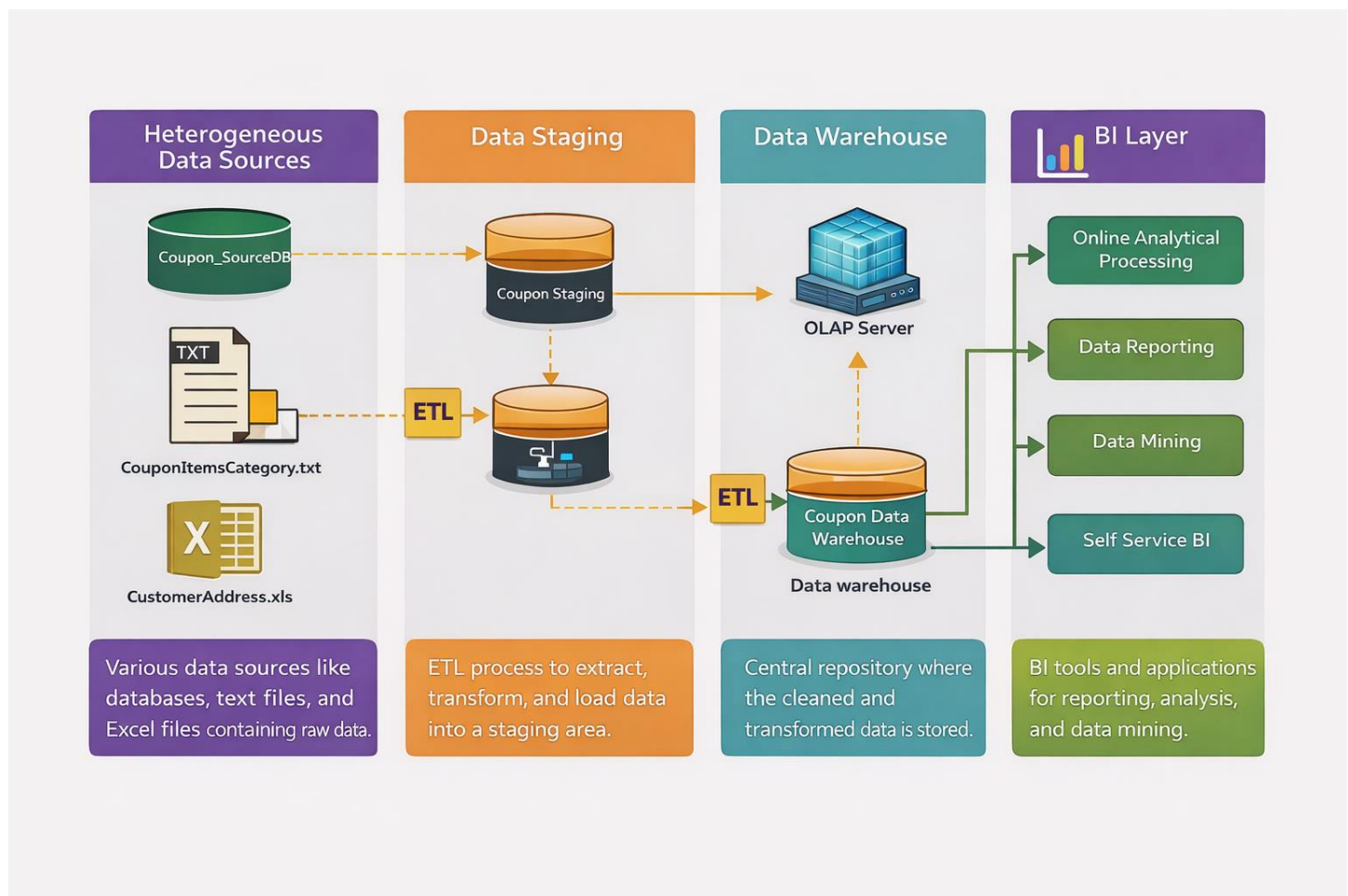
years_experience

hospital_branch

SCD Type 1 (Overwrite) - Only the latest value is required.

Applied for – phone_number

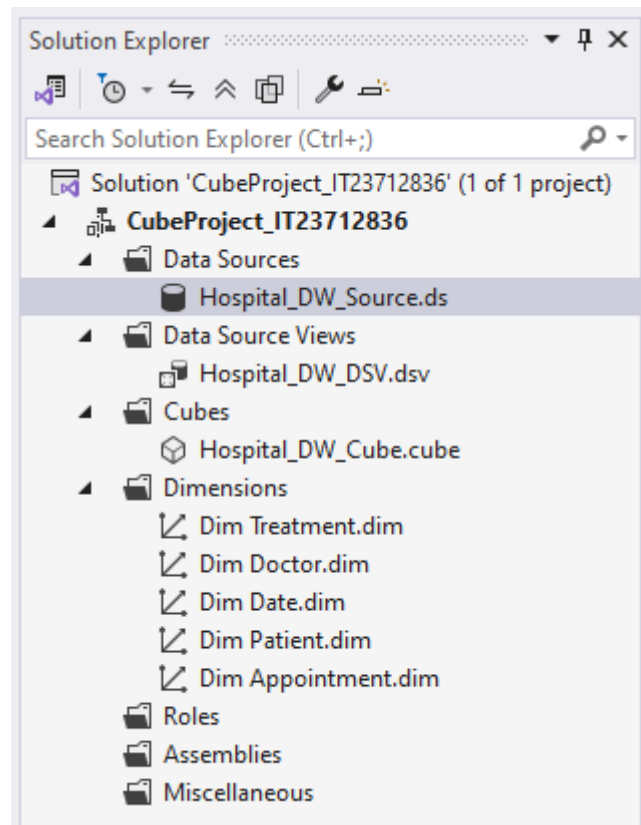
Email



The Data warehouse created is used as a source to create an OLAP Cube which helps to carry out efficient business intelligence principles improving the performance of business.

2)SSAS Cube implementation

SQL Server Data Tools (SSDT) was used to create an Analysis Services Multidimensional Project in order to develop the SSAS cube using the Hospital_DW data warehouse as the source. The cube enables efficient analysis of hospital operations such as patient treatments, billing, and appointments.



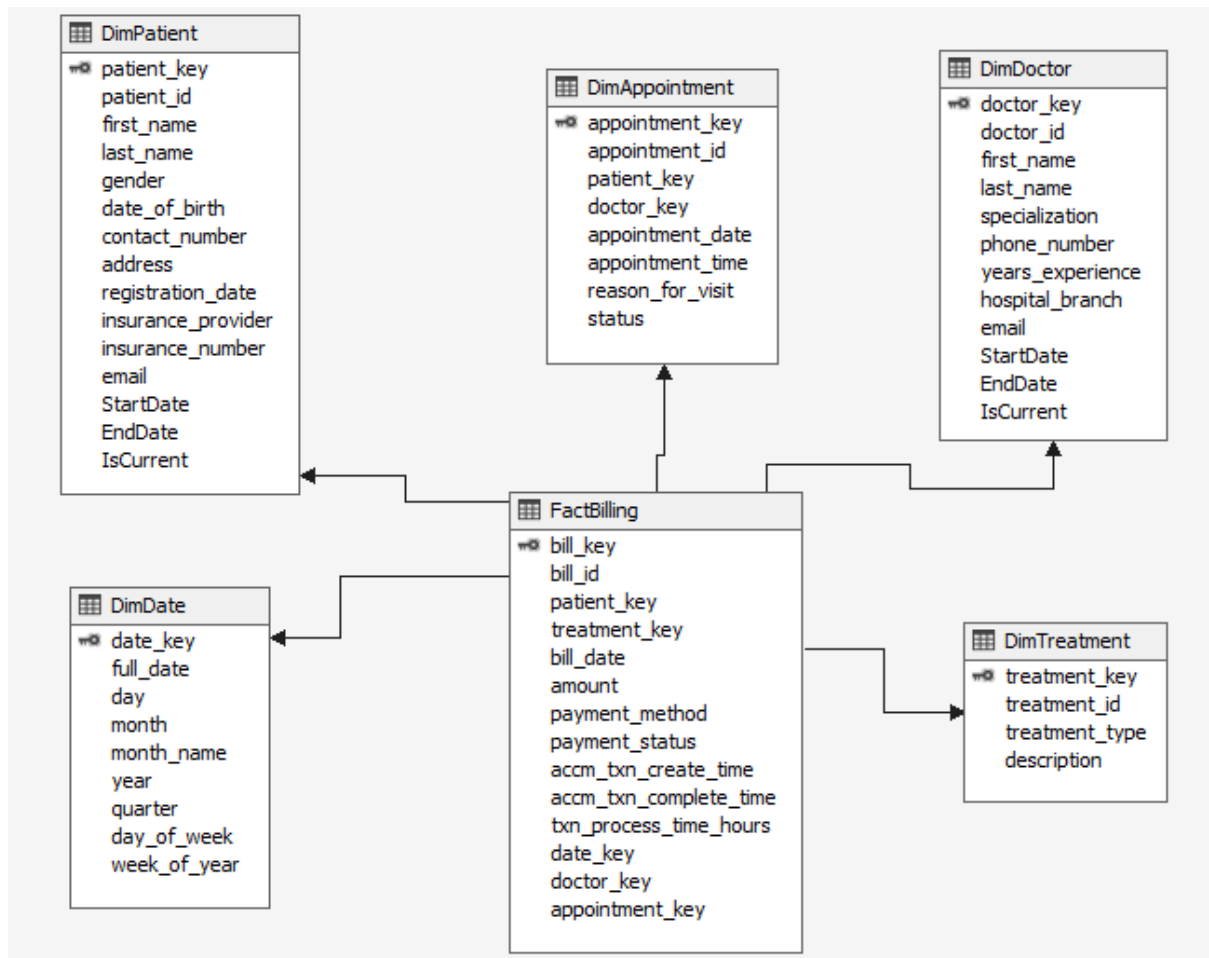
To create the SSAS cube, the following main steps were carried out:

- Data Source Creation (Hospital_DW_Source)
- Data Source View Creation (Hospital_DW_DSV)
- Data Source View Creation (Hospital_DW_Cube)

2.1 Data Source View

After creating the data source, a Data Source View (DSV_Hospital_DW) was developed using the Hospital_DW database.

The relationships between the fact table and dimension tables were defined using primary keys and foreign keys. Surrogate keys from dimension tables were linked with the corresponding keys in the FactBilling table to ensure proper integration of data for analysis.

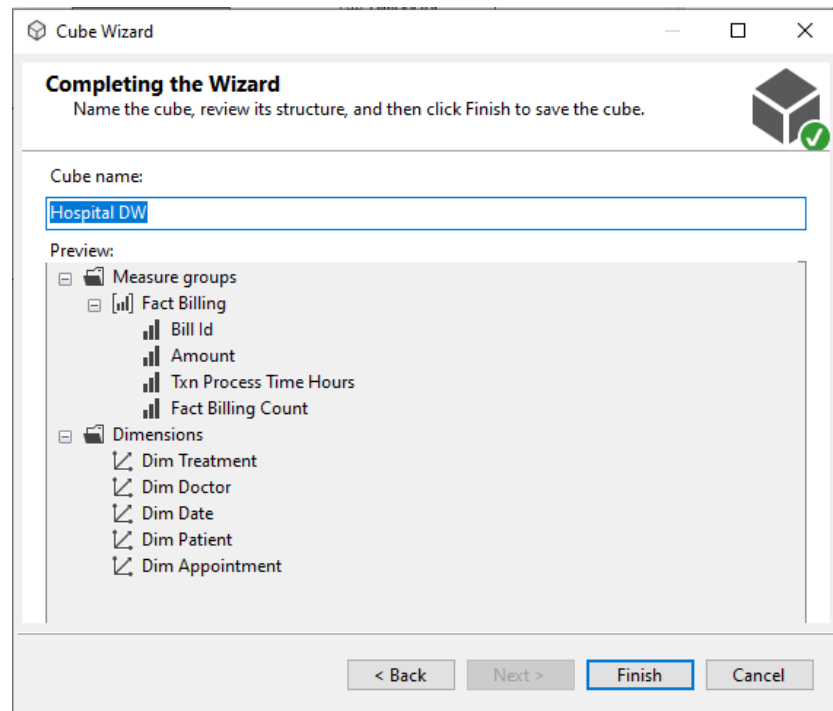


2.2 Cube Structure

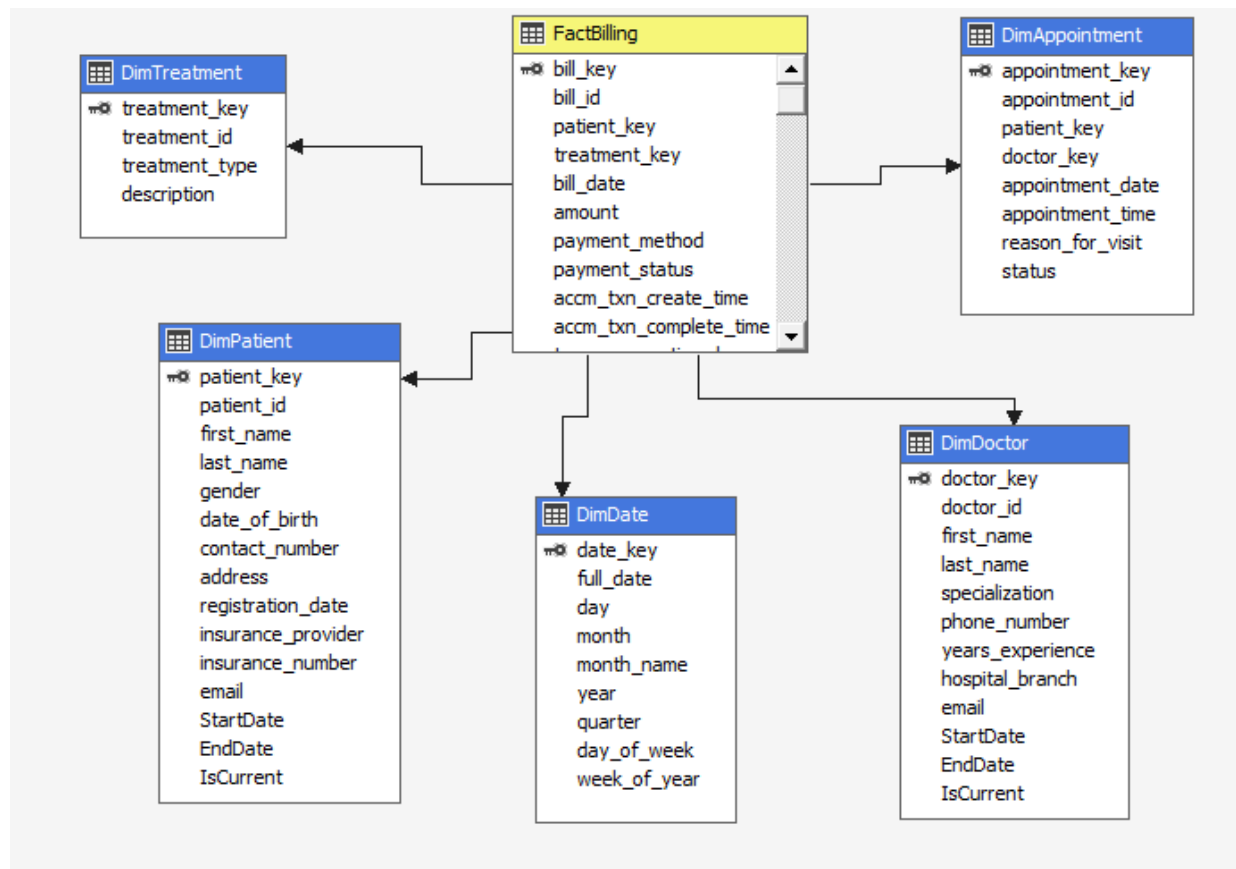
Finally, the cube structure (Cube_Hospital) was created using the data source view. The required dimensions and fact table were selected through the cube wizard to build the analytical model.

All dimension tables are directly connected to the FactBilling table, forming a star schema structure. The DimDate dimension is used to support time-based analysis, while other dimensions allow analysis based on patients, doctors, treatments, and appointments.

All dimension tables are directly connected to the FactBilling table, forming a star schema structure

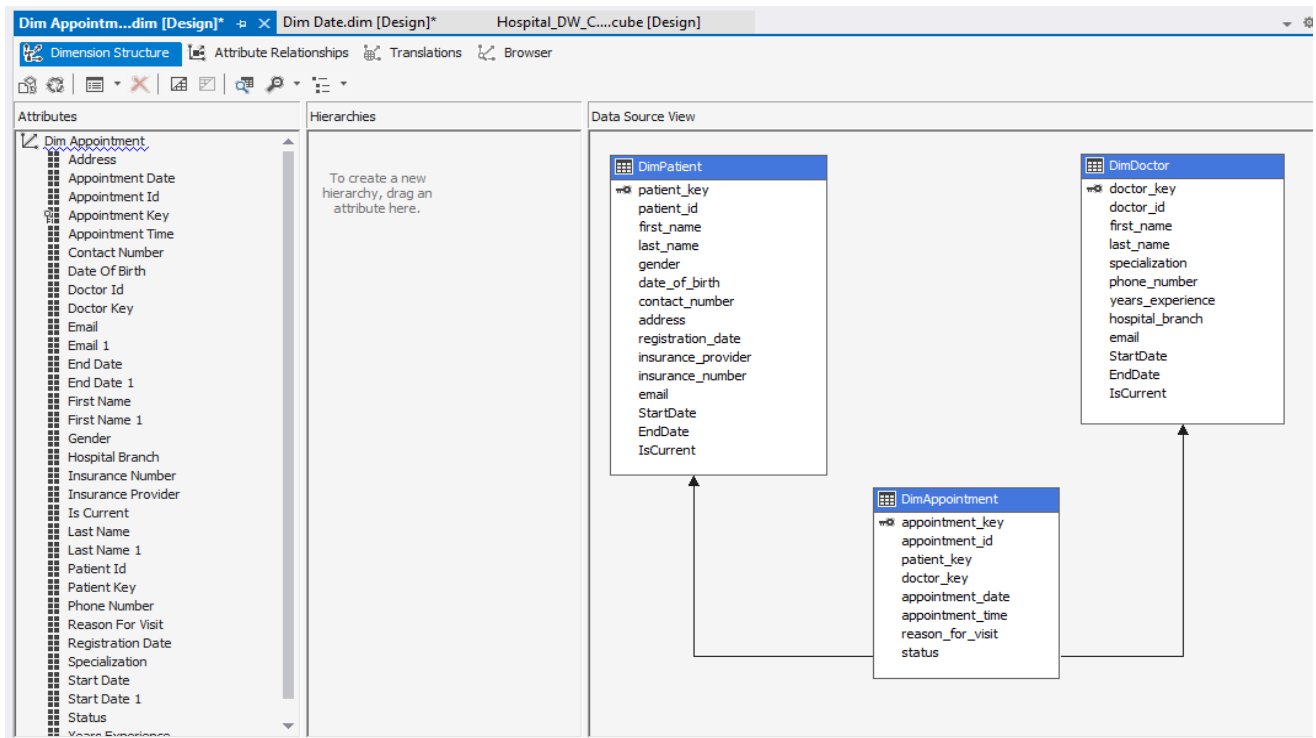


The final cube consists of one fact table and five main dimensions:



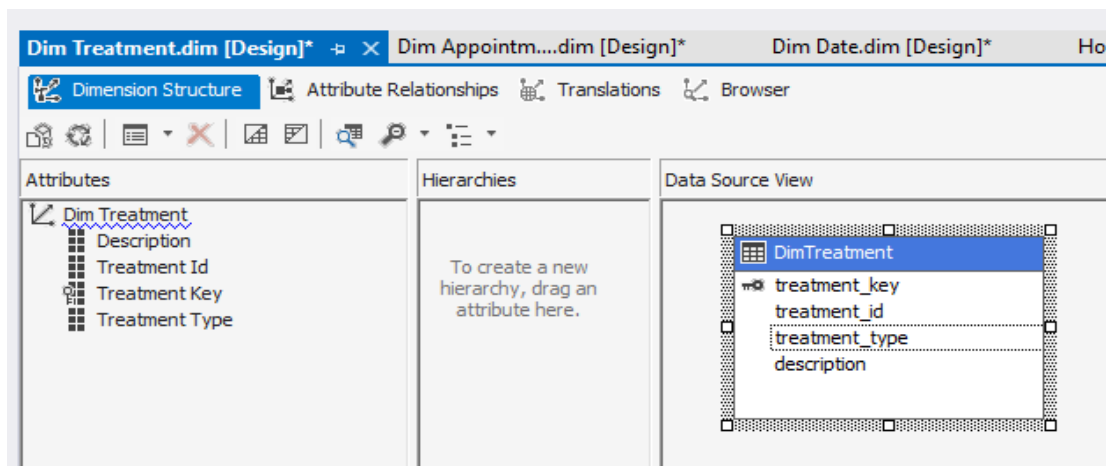
Patient, Doctor, Appointment Dimension

The cube uses separate dimensions for Patient, Doctor, and Appointment, each directly connected to the FactBilling table forming a star schema



Dimension – Treatment

The Treatment dimension contains information about medical treatments provided in the system. It is used to analyze the types of treatments associated with patient appointments and billing records.



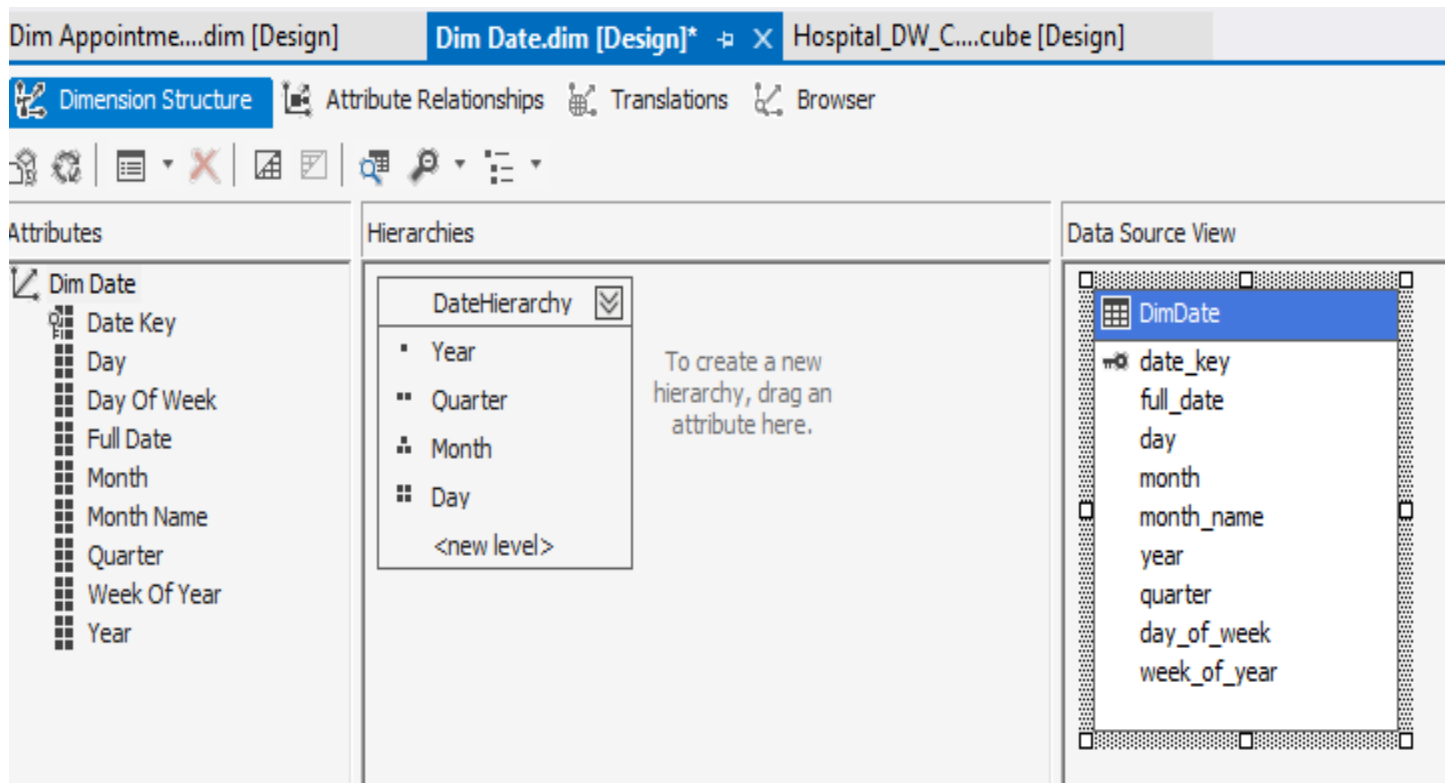
2.3 HIERARCHY

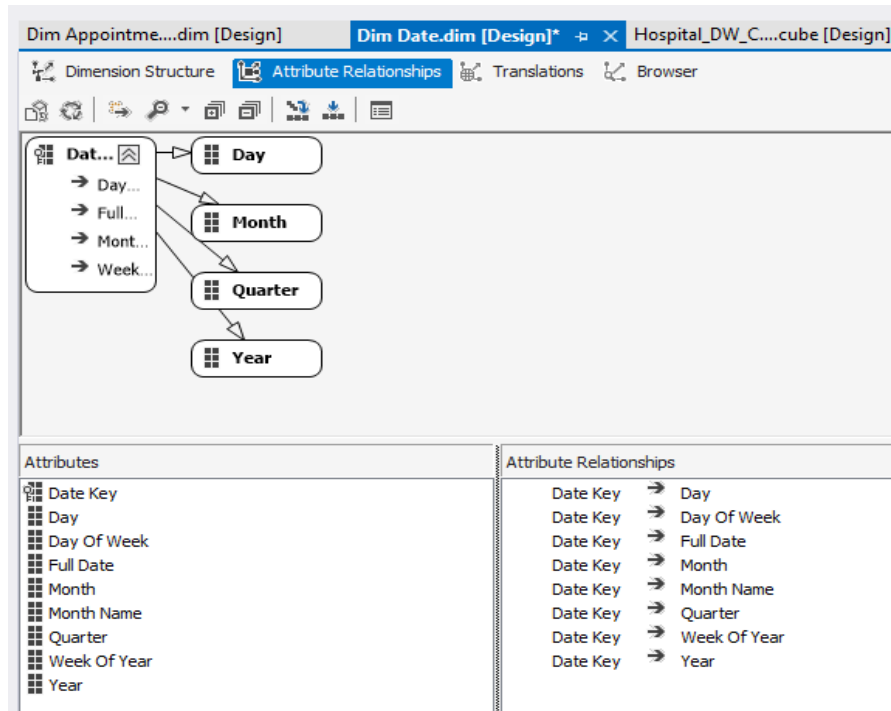
The Star schema in the data warehouse includes hierarchical relationships across several dimensions.

Hierarchies are created in DimDate, DimAppointment, and DimTreatment, mainly based on time details, doctor information, and patient attributes such as location or gender. These hierarchies support drill-down and roll-up operations, allowing efficient analysis from detailed medical records to summarized reports. They are defined within the Dimension Structure of each respective dimension.

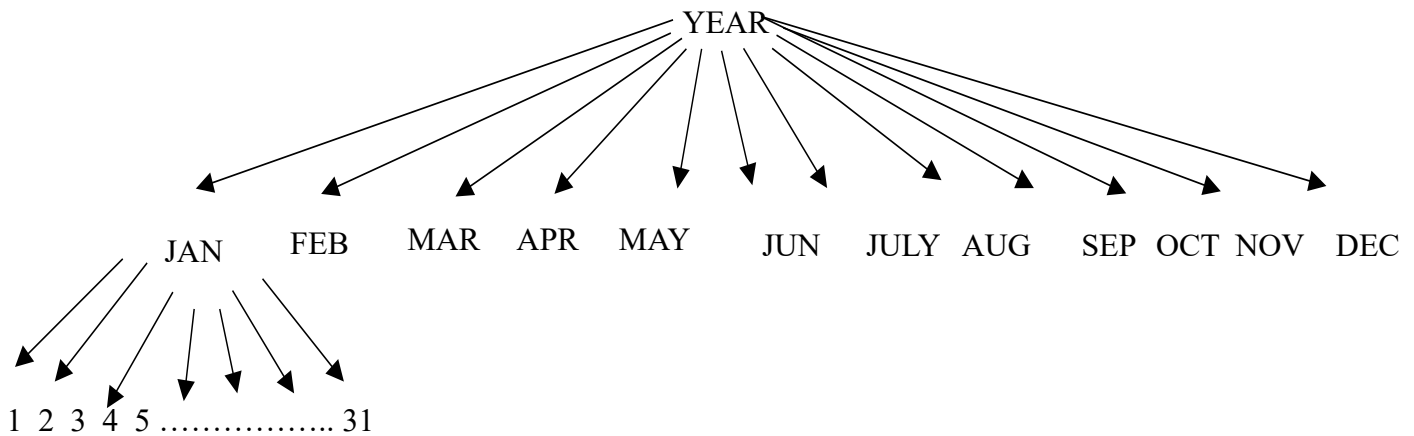
Date Hierarchy

The Date dimension was created to support time-based analysis in the cube. It enables users to analyze data across different time periods such as year, quarter, month, and day. This dimension is linked to the fact table to support drill-down and roll-up operations.





As per requirement Year , Month , Date wise hierarchy is created to analyze data in a more detailed manner with respect to date



2.4 KPI Implemented in the Cube

Key Performance Indicators were implemented in the SSAS cube to evaluate hospital performance. KPIs provide a visual representation of business performance using measures, targets (goals), and status indicators

These KPIs help analyze revenue generation, doctor workload, and payment efficiency in the hospital system.

1) Revenue KPI

Purpose	To measure total hospital revenue and compare it against a predefined financial target.
Measure Used	[Measures].[Amount] (Total billing revenue)
Goal	Fixed target value - 3 000 000
Status Logic	Green → Revenue $\geq$ 100% of target Yellow → Revenue $\geq$ 80% of target Red → Revenue $<$ 80% of target

The screenshot displays the SSDT interface for configuring a KPI. The main window shows the 'KPIs' tab with the following configuration:

- Name:** Revenue KPI
- Associated measure group:** Fact Billing
- Value Expression:** [Measures].[Amount]
- Goal Expression:** 3000000
- Status:**
 - Status indicator:** Shapes
 - Status expression:**

```
Case
  When KPIValue("Revenue KPI") / KPIGoal("Revenue KPI") >= 1 Then 1
  When KPIValue("Revenue KPI") / KPIGoal("Revenue KPI") >= 0.8 Then 0
  Else -1
```
- Trend:**
 - Trend indicator:** Standard arrow
 - Trend expression:**

The left pane shows the 'KPI Organizer' with 'Revenue KPI' selected. The right pane shows the 'Solution Explorer' with the project structure and the 'Properties' window for the 'Revenue KPI' object.

2) Doctor Service Volume KPI

Purpose	To measure the number of bills/consultations handled by doctors, representing workload and service activity.
Measure Used	[Measures].[Fact Billing Count]
Goal	Fixed or adjustable target
Status Logic	Green → Revenue $\geq$ 100% of target Yellow → Revenue $\geq$ 80% of target Red → Revenue $<$ 80% of target

The screenshot displays the SSDT KPI Designer interface for the 'Hospital_DW_Cube'. The main configuration pane shows the following details for the 'Doctor Service Volume KPI':

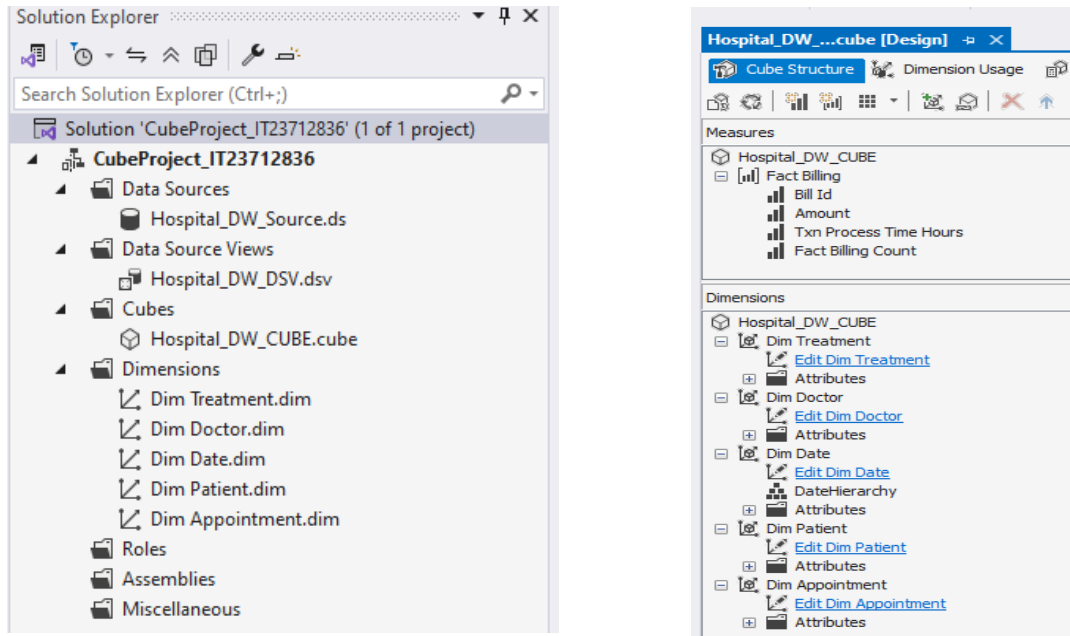
- Name:** Doctor Service Volume KPI
- Associated measure group:** Fact Billing
- Value Expression:** [Measures].[Bill Count]
- Goal Expression:** 500
- Status:**
 - Status indicator:** Shapes
 - Status expression:**

```
Case
when KPIValue("Doctor Service Volume KPI") / KPIGoal("Doctor Service Volume KPI") >= 1
Then 1
when KPIValue("Doctor Service Volume KPI") / KPIGoal("Doctor Service Volume KPI") >= 0.8
Then 2
else 0
```
- Trend:**
 - Trend indicator:** Standard arrow
 - Trend expression:** (Empty)

The left pane shows the 'KPI Organizer' with 'Doctor Service Volume KPI' selected. The right pane shows the 'Solution Explorer' with the project structure for 'CubeProject_IT23712836', including 'Data Sources', 'Data Source Views', 'Cubes', 'Dimensions', 'Roles', and 'Assemblies'. The bottom status bar shows 'Ready' and 'Add to Source Control' and 'Select Repository' buttons.

2) Demonstration of OLAP operations

Once the cube was deployed successfully, it was accessed through SQL Server Analysis Services. The cube was connected to Excel using the Data tab, and Pivot Tables were created to demonstrate OLAP operations.



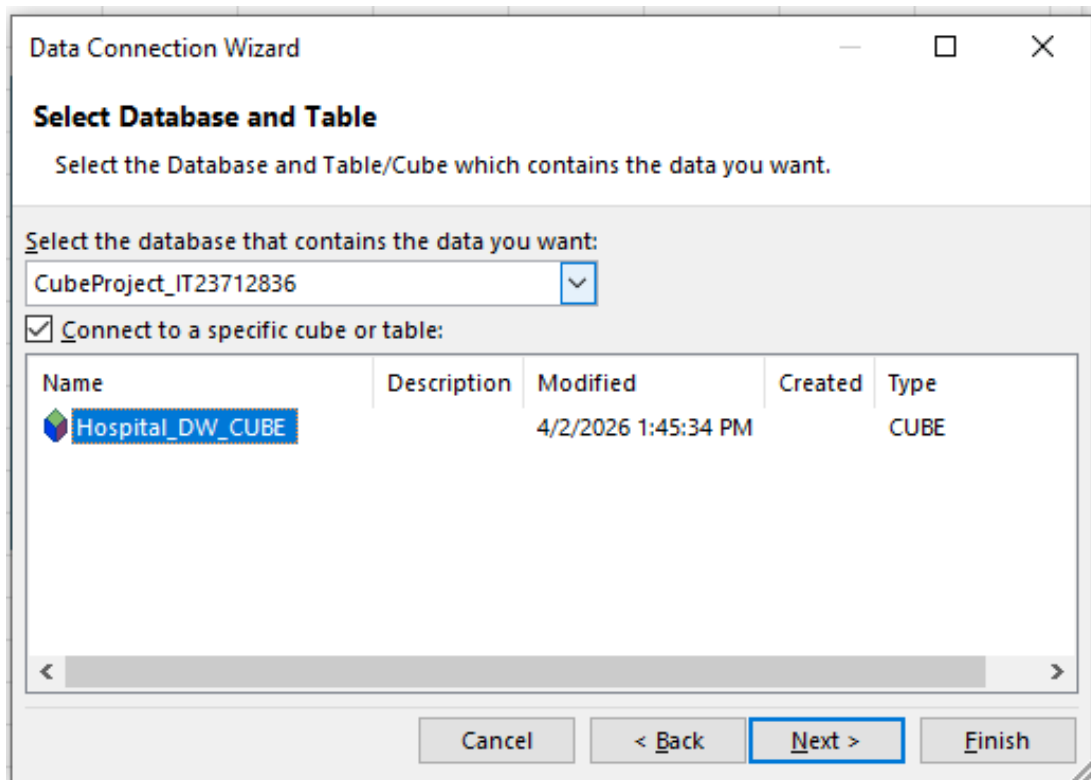
This Cube Structure can be used to browse data in SQL Server Analysis Service where queries can be passed as MDX query for visualization processes.

Appointment Id	Last Name	Contact Number	Reason For Visit	Hospital Branch	Status	Amount
1	Wilson	702.675.8848	Checkup	Kandy	Completed	5452
1000	Gutierrez	743-818-0486x3737	Emergency	Jaffna	Cancelled	19568
1003	Rivas	001-390-831-882...	Follow-up	Colombo	Scheduled	189822
101	Kelley	001-891-598-392...	Follow-up	Kandy	Cancelled	52698
1010	Owens	-1904	Follow-up	Galle	Scheduled	73218
1012	Walker	882.399.5239x9087	Follow-up	Kandy	Completed	91204
1015	Owens	350-351-8620x8102	Emergency	Galle	Completed	15176
1016	Henry	(772)370-4156	Consultation	Jaffna	Cancelled	48732
1018	Smith	(565)778-4788	Checkup	Colombo	Scheduled	82812
1019	Spears	842.468.6539	Follow-up	Colombo	Scheduled	192604
102	Wilson	(486)825-5785x582	Checkup	Kandy	Completed	45136
1020	Johnson	7375215304	Follow-up	Kandy	Completed	152664
1021	Buckley	(769)796-1579x9...	Emergency	Colombo	Scheduled	46548
1022	White	3022771356	Emergency	Kandy	Cancelled	4904
1026	Anthony	461.850.7968x357	Emergency	Jaffna	Cancelled	93840
1028	Jones	6357284046	Follow-up	Jaffna	Scheduled	61324
1029	Gonzalez	+1-559-950-7355...	Consultation	Galle	Cancelled	99096
1032	Jackson	822.435.7999	Consultation	Galle	Completed	101640
1033	Wells	5077140260	Emergency	Colombo	Scheduled	202042

There are some main OLAP Operations demonstrated through Excel visualizations

- Slice Operations
- Dice Operations
- Pivot Operations
- Drill-Down Operations
- Roll-Up Operations

To demonstrate all of these operations , Excel workbook was connected to the cube via Data tab. Data from Cube is considered as the data source and is obtained via Analysis services.



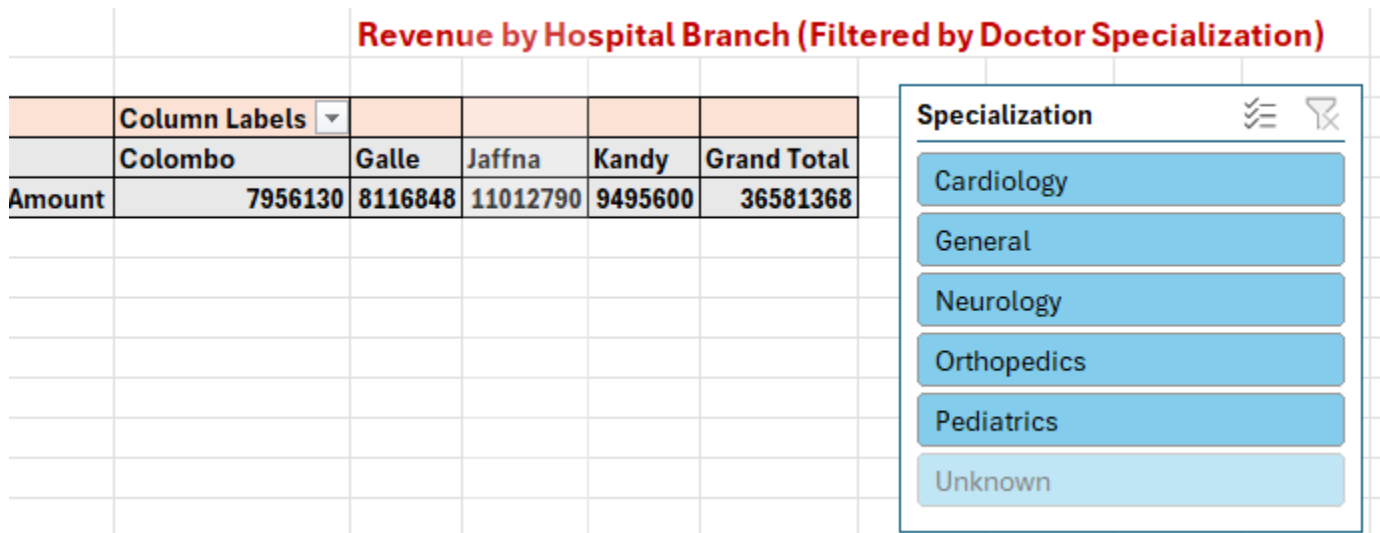
For all the operations Pivot charts and tables can be generated using Pivot Chart fields which was obtained from the Cube accessed via Analysis Service of Data tab.

3.1 SLICE Operation

Slice operation is performed by selecting a single dimension from the cube. In this scenario, Doctor Specialization was used as a slicer to filter and analyze revenue (Amount) across different hospital branches.

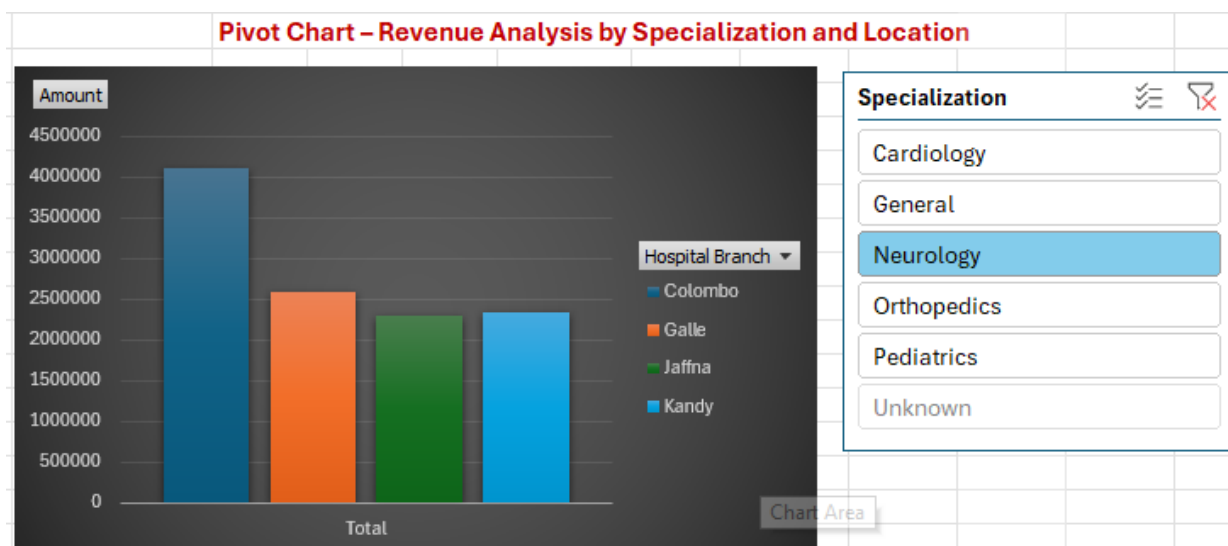
Revenue by Hospital Branch (Filtered by Doctor Specialization)

A Pivot Table was created using Amount as the measure and Hospital Branch as the dimension. A slicer was applied using Doctor Specialization to filter and analyze revenue based on different medical categories.

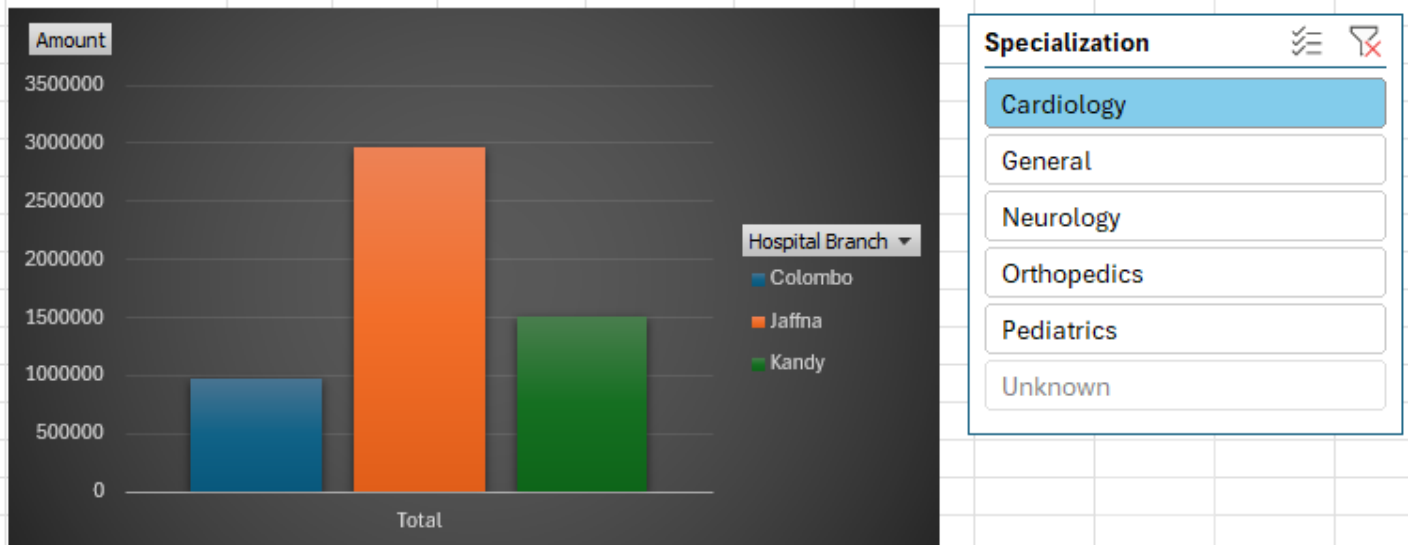


Pivot Chart – Revenue Analysis by Specialization and Location

A Pivot Chart was generated from the Pivot Table to visually represent revenue across hospital branches. The chart dynamically updates based on the selected Doctor Specialization from the slicer.



Pivot Chart – Revenue Analysis by Specialization and Location



3.2 Dice Operation

Dice operation is performed by selecting multiple dimensions from the cube to analyze data under specific conditions. In this scenario, Doctor Specialization and Year were used to filter the Amount measure, allowing analysis of revenue across different specializations and time periods.

Revenue by Hospital Branch (Filtered by Specialization and Year)

Revenue by Hospital Branch (Filtered by Specialization and Year)					
Column Labels					
	Colombo	Galle	Jaffna	Kandy	Grand Total
Amount	7956130	8116848	11012790	9495600	36581368

Years Experience

23

24

25

26

27

28

3

30

Specialization

Cardiology

General

Neurology

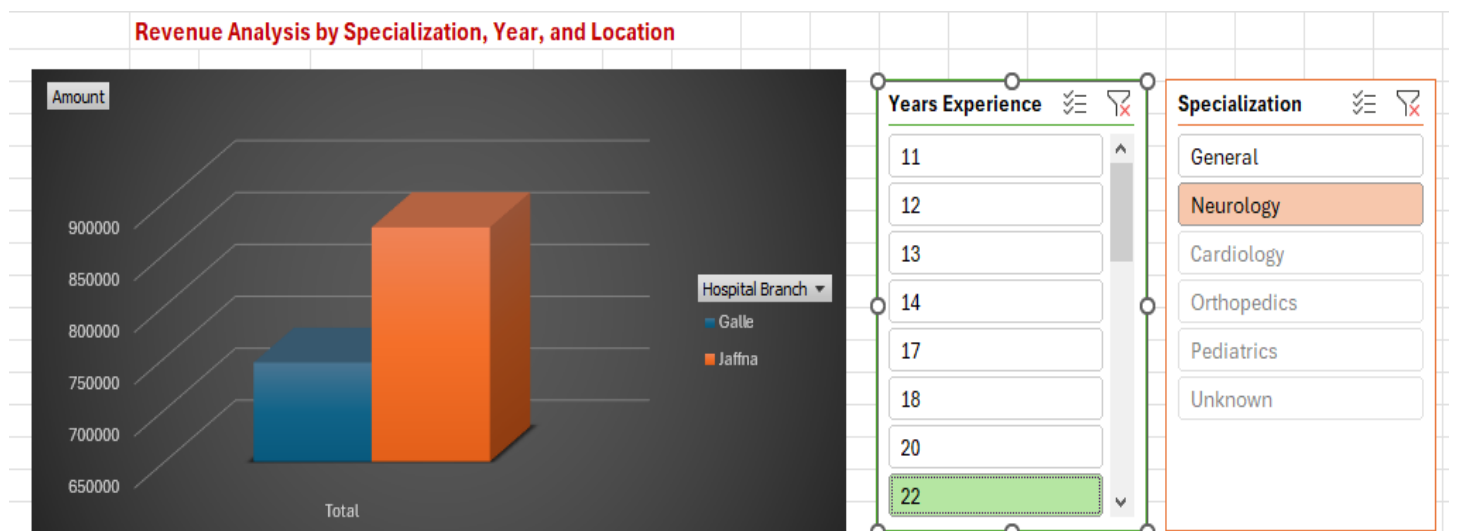
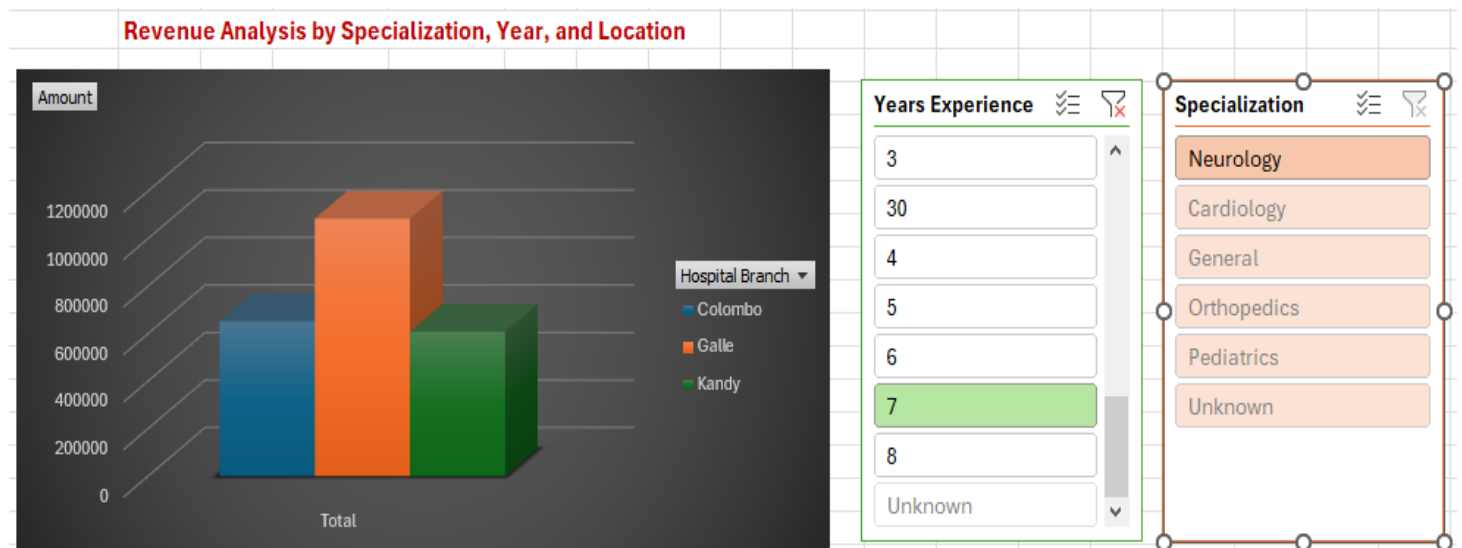
Orthopedics

Pediatrics

Unknown

Revenue Analysis by Specialization, Year, and Location

A Pivot Table was created using Amount as the measure and Hospital Branch as the dimension. Two slicers, Doctor Specialization and Year, were used to filter data and perform multi-dimensional analysis.



3.3 Pivot Operation

Pivot operation is used to rearrange data by rotating rows and columns to view information from different perspectives. In this scenario, Hospital Branch and Year were alternated between rows and columns to analyze revenue (Amount) in multiple ways.

Revenue by Hospital Branch and Year (Initial View)

The Pivot Table displays revenue across hospital branches for different years, with Hospital Branch in rows and Year in columns.

Revenue by Hospital Branch and Year (Initial View)					
Amount	Column Labels				
Row Labels	2020	2021	2022	2023	Grand Total
Colombo	2548270	2531960	2093078	782822	7956130
Galle	2086666	2642120	2804384	583678	8116848
Jaffna	3491172	3190212	3290868	1040538	11012790
Kandy	2803232	2770344	3378714	543310	9495600
Grand Total	10929340	11134636	11567044	2950348	36581368

Drag fields between areas below:

Filters	Columns
	Year
Rows	Values
Hospital Branch	Amount

Revenue by Year and Hospital Branch (Pivoted View)

The rows and columns were interchanged to provide a different perspective of the same data, demonstrating the Pivot operation.

Revenue by Year and Hospital Branch (Pivoted View)					
Amount	Column Labels ▾				
Row Labels ▾	Colombo	Galle	Jaffna	Kandy	Grand Total
2020	2548270	2086666	3491172	2803232	10929340
2021	2531960	2642120	3190212	2770344	11134636
2022	2093078	2804384	3290868	3378714	11567044
2023	782822	583678	1040538	543310	2950348
Grand Total	7956130	8116848	11012790	9495600	36581368

Drag fields between areas below:

Filters 	Columns Hospital Branch ▾
Rows Year ▾	Values Amount ▾

3.4 Drill-Down and Roll-Up Operation

Drill-down and roll-up operations are used to navigate through different levels of data within a hierarchy. Drill-down allows users to move from summarized data to more detailed levels, while roll-up aggregates detailed data into higher-level summaries.

Drill-Down Operation

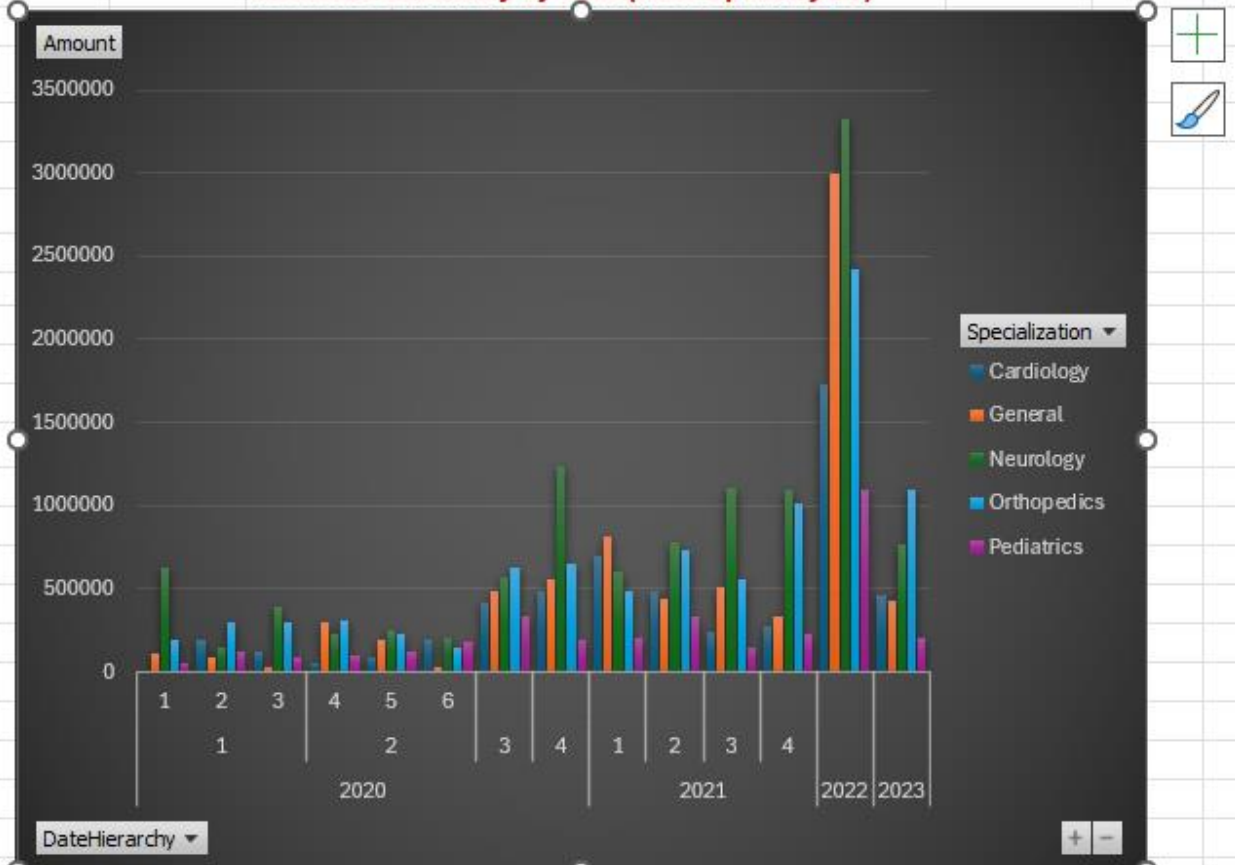
Drill-Down Operation using Date Hierarchy

In this scenario, drill-down was performed using the Date hierarchy by expanding data from Year level to Month level. This allows detailed analysis of revenue (Amount) for each month within a selected year.

Revenue Analysis by Date Hierarchy (Drill-Down & Roll-Up)

Amount	Column Labels					
Row Labels	Cardiology	General	Neurology	Orthopedics	Pediatrics	Grand Total
2020						
1						
1		107776	631380	194306	53968	987430
2	193908	93322	144518	303372	126378	861498
3	125174	25258	392332	293278	88534	924576
2						
4	53974	301224	224218	306482	96730	982628
5	83446	195306	248744	229976	121346	878818
6	188142	24124	209538	143180	184054	749038
3	417148	481100	573668	622942	330970	2425828
4	491884	553916	1231440	648424	193860	3119524
2021						
1	699614	820368	602468	491724	209132	2823306
2	492230	444730	785316	731150	331590	2785016
3	245292	514032	1104602	561856	148438	2574220
4	274436	336632	1096116	1012156	232754	2952094
2022	1725378	2995072	3320680	2424414	1101500	11567044
2023	462222	422824	765118	1093486	206698	2950348
Grand Total	5452848	7315684	11330138	9056746	3425952	36581368

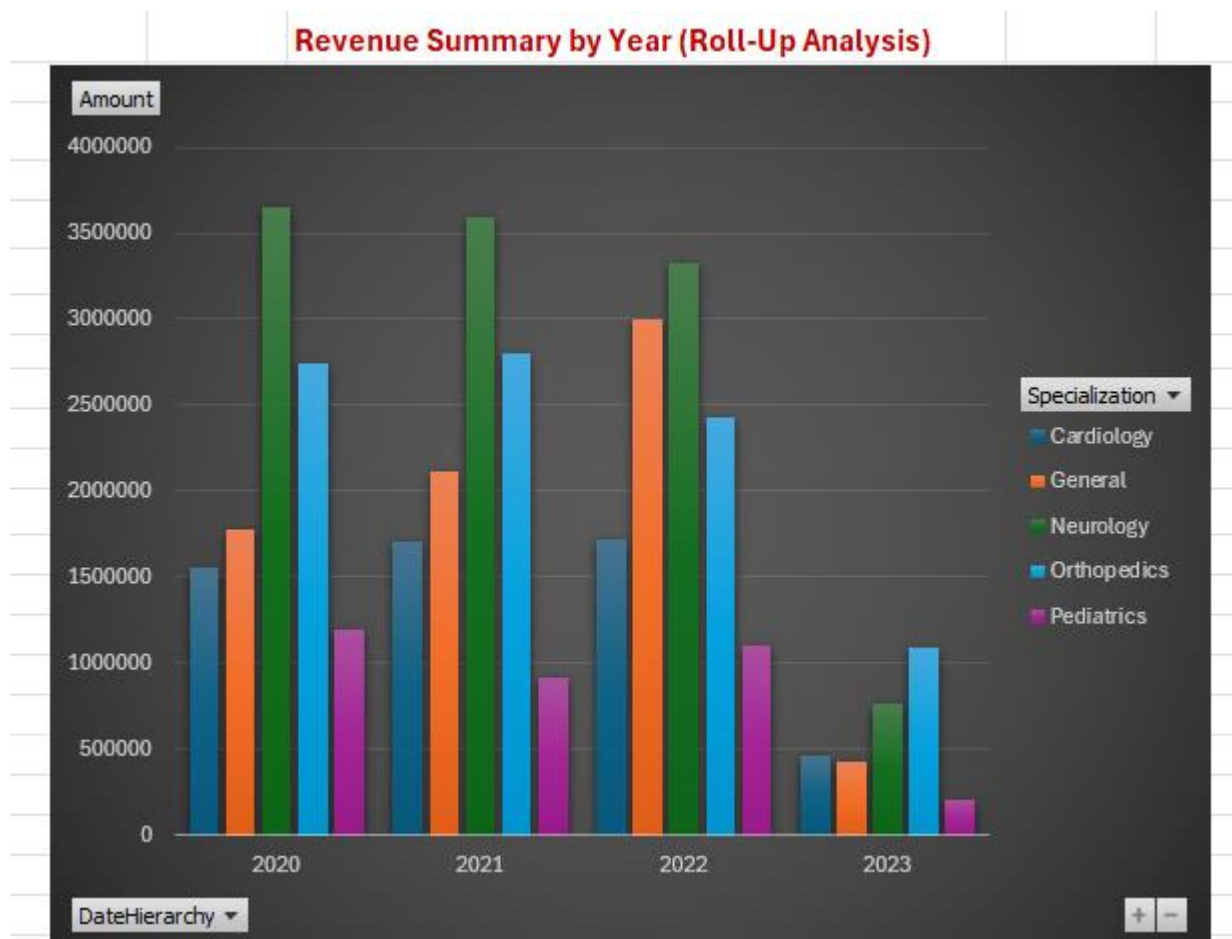
Revenue Summary by Year (Roll-Up Analysis)



Roll-Up Operation using Date Hierarchy

Roll-up was performed by collapsing the Month level back to Year level. This aggregates detailed monthly data into yearly totals, providing a summarized view of revenue.

Revenue Analysis by Date Hierarchy (Drill-Down & Roll-Up)						
Amount	Column Labels					
Row Labels	Cardiology	General	Neurology	Orthopedics	Pediatrics	Grand Total
2020	1553676	1782026	3655838	2741960	1195840	10929340
2021	1711572	2115762	3588502	2796886	921914	11134636
2022	1725378	2995072	3320680	2424414	1101500	11567044
2023	462222	422824	765118	1093486	206698	2950348
Grand Total	5452848	7315684	11330138	9056746	3425952	36581368



PivotChart Fields

Choose fields to add to report:

Search

Dim Date

- DateHierarchy
- More Fields

Dim Doctor

Drag fields between areas below:

Filters

Axis (Categories)

DateHierarchy

Legend (Series)

Specialization

Values

Amount

Page | 24

4) PowerBI Reports

Power BI is a business intelligence tool used to create interactive dashboards and reports by connecting to various data sources such as SQL Server Analysis Services (SSAS). In this project, Power BI Desktop was used to connect to the developed OLAP cube and generate interactive visualizations for analytical purposes.

The reports were designed to provide insights into hospital operations, including revenue generation, patient distribution, appointment trends, and doctor performance. After designing the reports in Power BI Desktop, they can be published to Power BI Service for online access and sharing.

4.1 Report 1 – Revenue Structure Dashboard (Matrix-Based)

This report focuses on presenting structured and detailed tabular data using a **Matrix visual**, which allows grouping by rows and columns.

[Back to report](#)

Specialization	Colombo	Galle	Jaffna	Kandy	Total
Neurology	4,110,052.00	2,587,416.00	2,290,020.00	2,342,650.00	11,330,138.00
Orthopedics	2,327,688.00	1,843,586.00	2,702,496.00	2,182,976.00	9,056,746.00
General		2,973,674.00	884,224.00	3,457,786.00	7,315,684.00
Cardiology	977,476.00		2,963,184.00	1,512,188.00	5,452,848.00
Pediatrics	540,914.00	712,172.00	2,172,866.00		3,425,952.00
Total	7,956,130.00	8,116,848.00	11,012,790.00	9,495,600.00	36,581,368.00

The matrix includes:

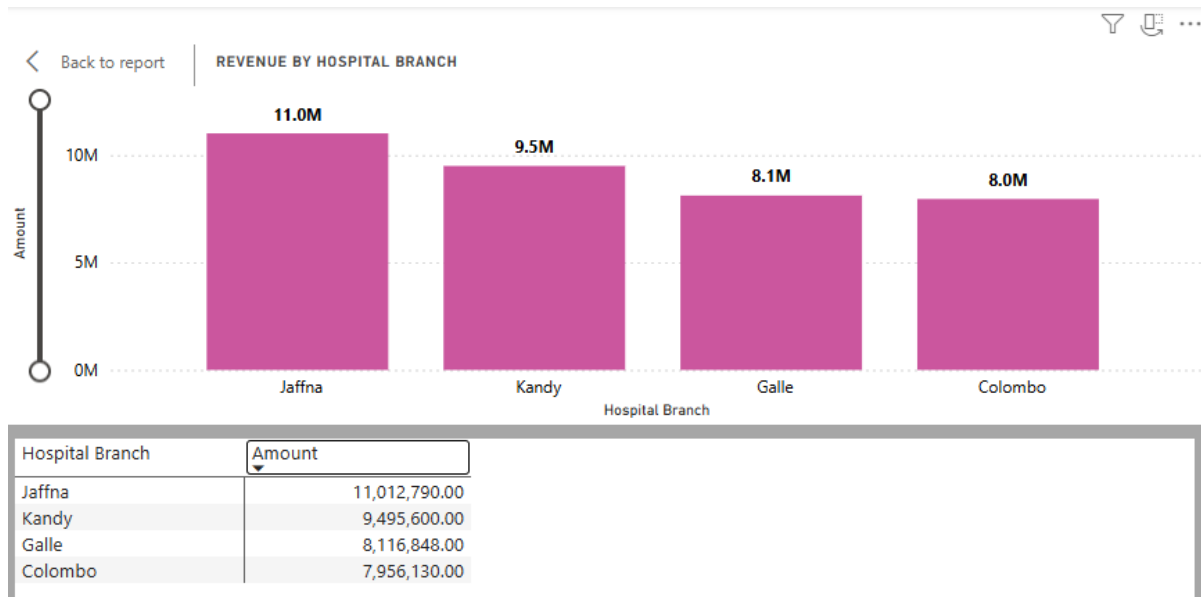
- Revenue (Amount) as the primary measure
- Doctor Specialization as row grouping
- Hospital Branch as column grouping
- Total revenue aggregation for overall comparison

This structure allows users to clearly compare how each specialization contributes to revenue across different branches.

Supporting Visuals

To enhance analytical understanding, additional visuals were created alongside the matrix.

1. Bar Chart – Revenue by Hospital Branch



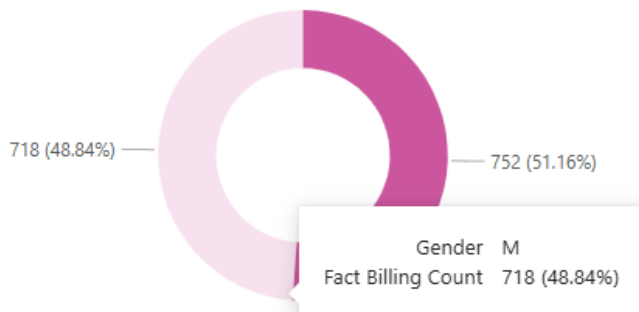
A bar chart was created to visualize total revenue generated by each hospital branch.

- X-axis → Hospital Branch
- Y-axis → Revenue (Amount)

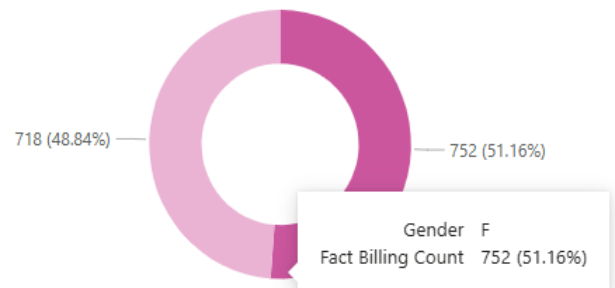
This visualization helps identify high-performing branches and compare revenue across locations.

2. Donut Chart – Revenue Distribution by Specialization

APPOINTMENT DISTRIBUTION BY GENDER



APPOINTMENT DISTRIBUTION BY GENDER

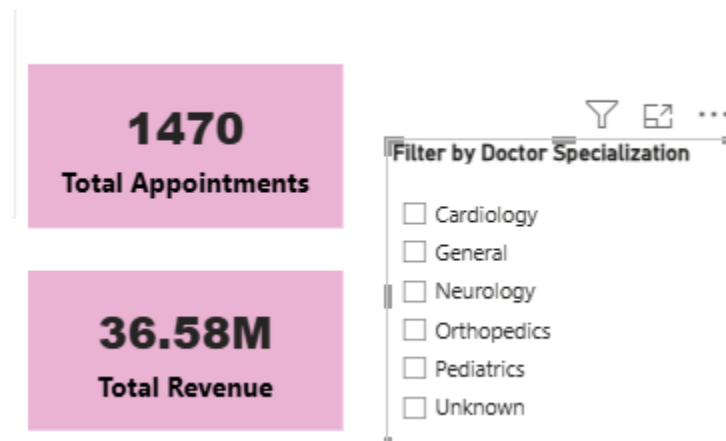


3. KPI Cards – Summary Indicators

Two KPI cards were added:

- **Total Revenue** → Displays overall revenue generated
- **Total Appointments** → Shows total number of appointments

These provide a quick high-level summary of system performance.



In addition, a slicer (Doctor Specialization) was included to enable interactive filtering. When a specialization is selected, both KPI values and all other visuals are dynamically updated to reflect the filtered data.

This allows users to quickly analyze how revenue and appointment counts vary across different doctor specializations.

Final Dashboard View

All visuals (Matrix, Bar Chart, Pie Chart, KPI Cards, and Slicer) were combined into a single dashboard layout.

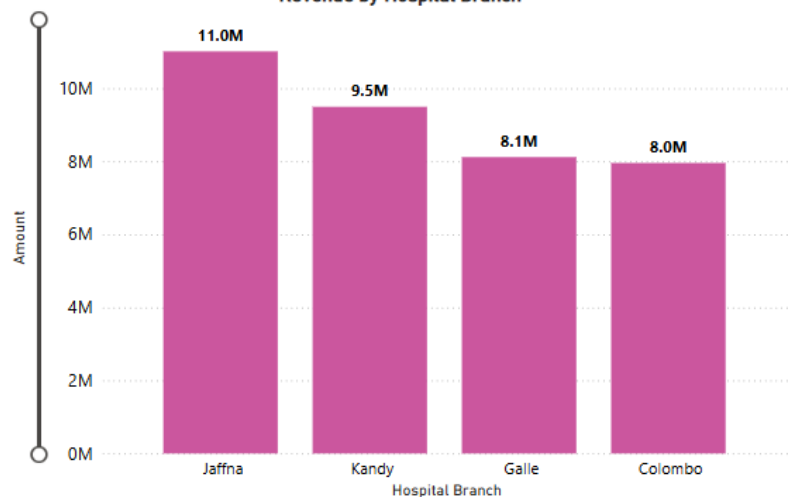
Each visual serves a different analytical purpose:

- Matrix → Detailed structured comparison across dimensions
- Bar Chart → Clear comparison of branch performance
- Pie Chart → Proportional contribution of specializations
- KPI Cards → Quick summary indicators

By combining these visuals, the dashboard provides both detailed and summarized insights into hospital revenue.

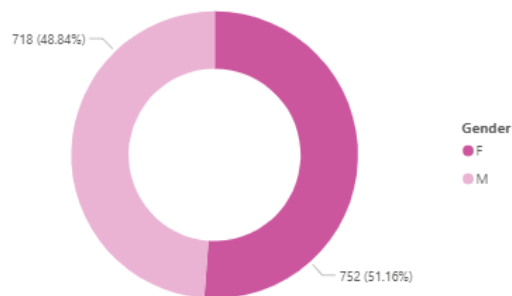
Hospital Revenue Analysis Dashboard

Revenue by Hospital Branch



Specialization	Colombo	Galle	Jaffna	Kandy	Total
Neurology	4,110,052.00	2,587,416.00	2,290,020.00	2,342,650.00	11,330,138.00
Orthopedics	2,327,688.00	1,843,586.00	2,702,496.00	2,182,976.00	9,056,746.00
General		2,973,674.00	884,224.00	3,457,786.00	7,315,684.00
Cardiology	977,476.00		2,963,184.00	1,512,188.00	5,452,848.00
Pediatrics	540,914.00	712,172.00	2,172,866.00		3,425,952.00
Total	7,956,130.00	8,116,848.00	11,012,790.00	9,495,600.00	36,581,368.00

Appointment Distribution by Gender



1470
Total Appointments

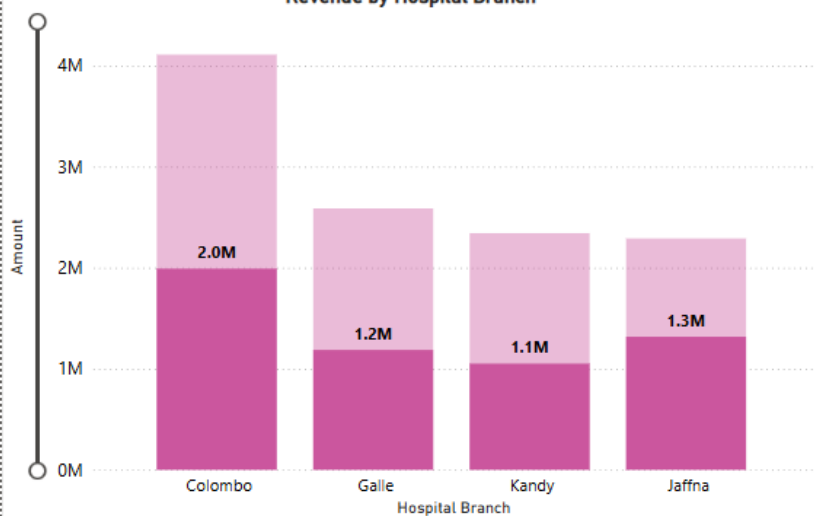
36.58M
Total Revenue

Filter by Doctor Specialization

- ☐ Cardiology
- ☐ General
- ☐ Neurology
- ☐ Orthopedics
- ☐ Pediatrics
- ☐ Unknown

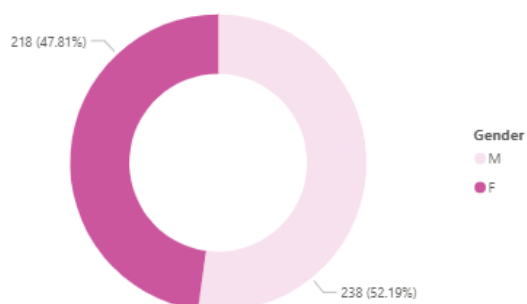
Hospital Revenue Analysis Dashboard

Revenue by Hospital Branch



Specialization	Colombo	Galle	Jaffna	Kandy	Total
Neurology	1,990,974.00	1,185,942.00	1,316,548.00	1,051,232.00	5,544,696.00
Total	1,990,974.00	1,185,942.00	1,316,548.00	1,051,232.00	5,544,696.00

Appointment Distribution by Gender



218
Total Appointments

5.54M
Total Revenue

Filter by Doctor Specialization

- ☐ Cardiology
- ☐ General
- ☒ Neurology
- ☐ Orthopedics
- ☐ Pediatrics
- ☐ Unknown

4.2 Report 2 - Operational Performance Dashboard (Interactive Analysis with Multiple Slicers)

This report focuses on analyzing overall hospital performance using an interactive dashboard. Unlike the matrix-based report, this dashboard is designed to support dynamic data exploration through multiple slicers and interactive visualizations.

The report combines time-based analysis, comparative insights, and distribution analysis to provide a comprehensive understanding of hospital operations.

KPI Cards – Summary Indicators

Two KPI cards were included:

- **Total Revenue** → Displays the total revenue generated across all hospital branches
- **Total Appointments** → Shows the total number of appointments recorded

These KPIs provide a quick overview of system performance and help users immediately understand the scale of operations before exploring detailed visuals.

Line Chart – Revenue Trend Over Time

A line chart was used to represent revenue trends across time.

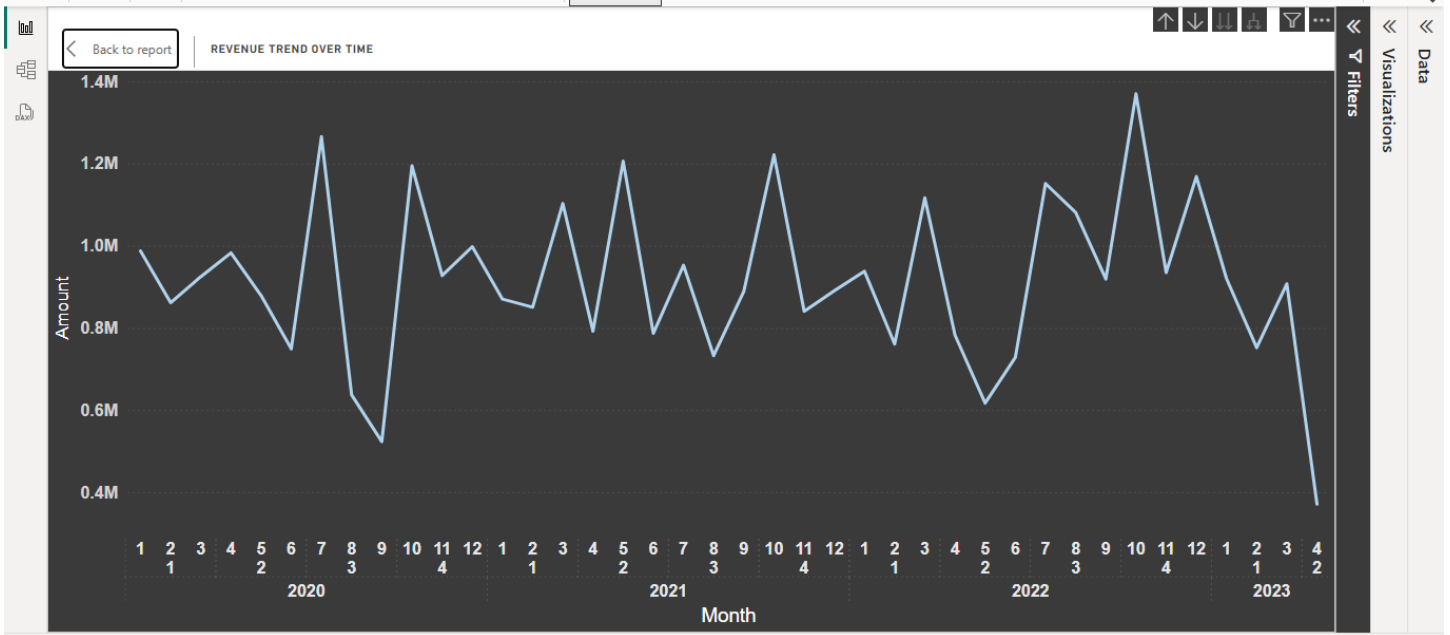
- X-axis → Month (Year-wise hierarchy)
- Y-axis → Revenue (Amount)

We are Using Line Charts Because :

A line chart is ideal for showing trends and patterns over time. It helps identify fluctuations, growth, and seasonal variations in revenue.

Analysis Insight :

Users can observe periods of high and low revenue, identify peak months, and analyze how revenue changes year by year.



Column Chart – Branch Performance Comparison

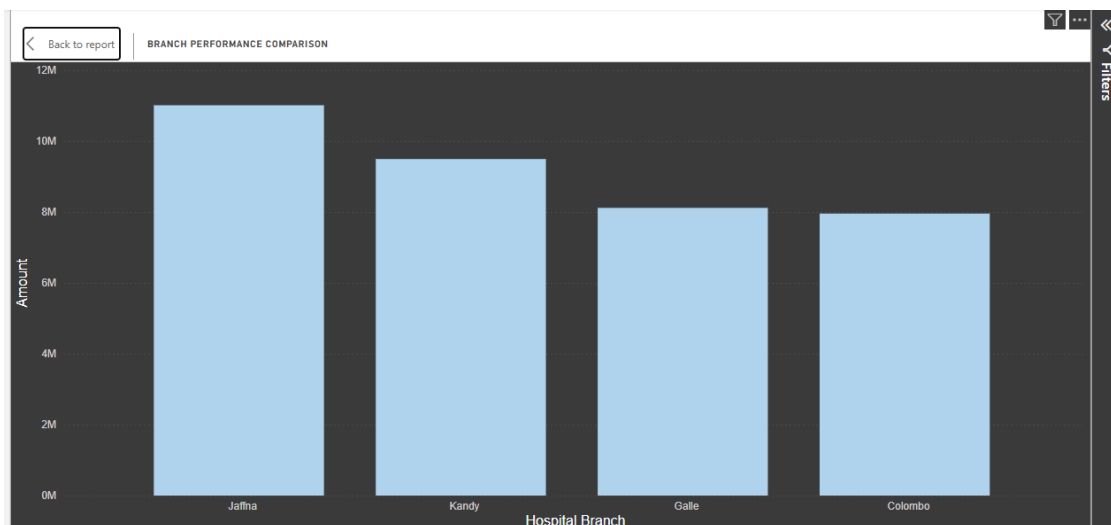
A column chart was created to compare revenue across hospital branches.

- X-axis → Hospital Branch
- Y-axis → Revenue (Amount)

We are Using Column chart Because :

Column charts are effective for comparing values across categories. This allows easy identification of which branch performs better.

Analysis Insight : Users can quickly identify high-performing branches (e.g., Jaffna) and lower-performing ones, supporting operational decisions.



Bar Chart – Revenue by Specialization

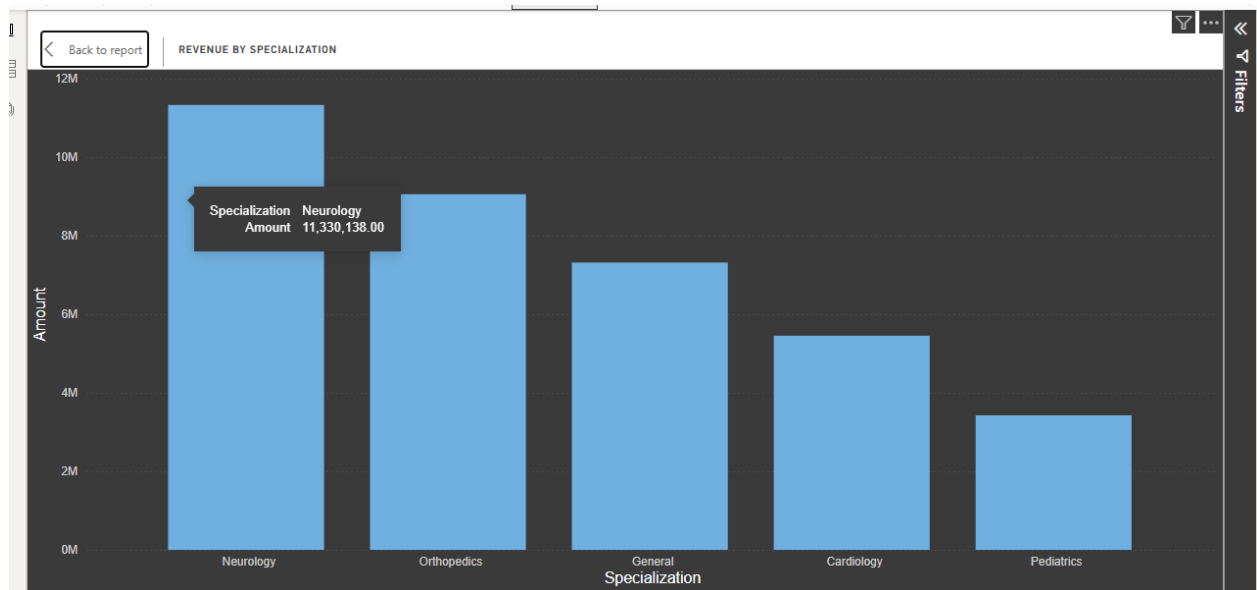
A bar chart was used to visualize revenue contribution by doctor specialization.

- X-axis → Revenue (Amount)
- Y-axis → Specialization

We are Using Bar chart Because :

Bar charts are suitable when category names are longer and allow better readability compared to column charts.

Analysis Insight : This helps determine which medical specializations generate the most revenue and which areas may require improvement.



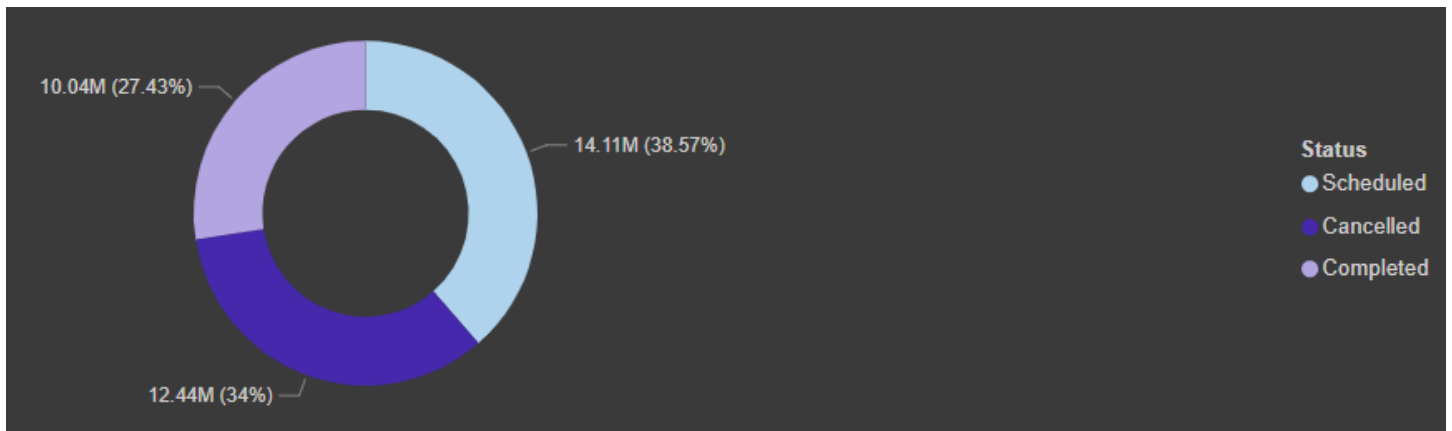
Donut Chart – Appointment Status Distribution

A donut chart was used to represent the distribution of appointments based on status.

- Legend → Appointment Status (Scheduled, Cancelled, Completed)
- Values → Appointment Count

Why we using donut Chart : Donut charts are ideal for showing proportional distribution of categories within a whole.

Analysis Insight: Users can understand the proportion of completed vs cancelled appointments, which is important for evaluating operational efficiency.

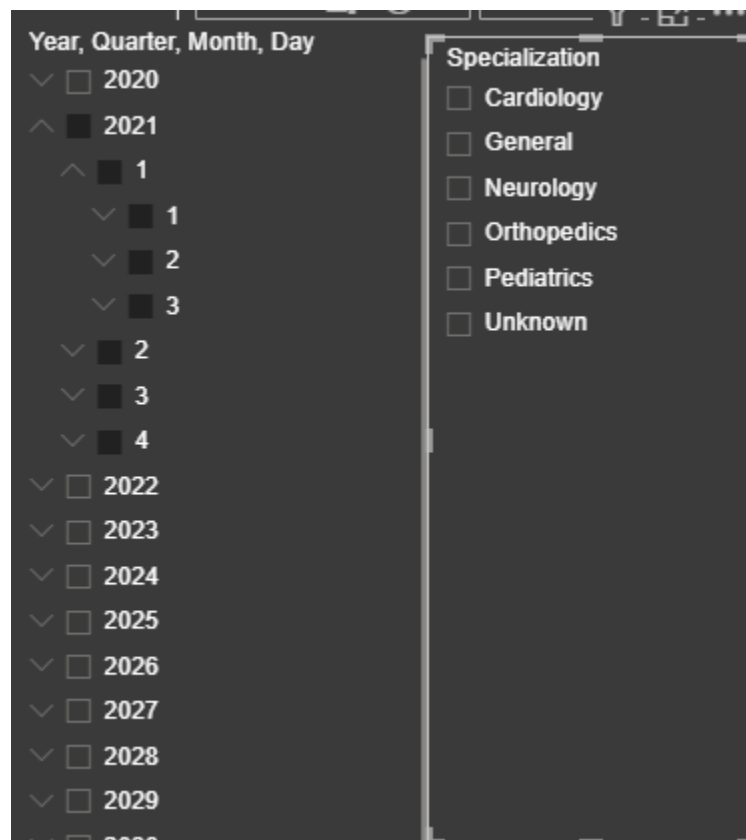


Slicers – Interactive Filtering

Slicers were used to enable dynamic filtering of the dashboard. Users can select values such as Date (Year/Month) and Doctor Specialization to control the data displayed across all visuals. When a selection is made, all charts and KPI cards update automatically, allowing focused analysis on specific data segments.

- Date (Year, Quarter, Month)
- Doctor Specialization

These slicers allow users to filter the dashboard based on specific conditions



Cascading Filter Behavior

The slicers were configured to work in a cascading manner. Selecting a value in one slicer (e.g., Year) restricts the available options in another slicer (e.g., Specialization). This ensures that only relevant and meaningful data combinations are displayed.

Impact on Visuals

All visuals are connected to the slicers. When a filter is applied, the line chart, branch chart, specialization chart, donut chart, and KPI cards update instantly, creating an interactive and user-driven analytical environment.

Final Dashboard View

This dashboard allows users to answer key business questions such as:

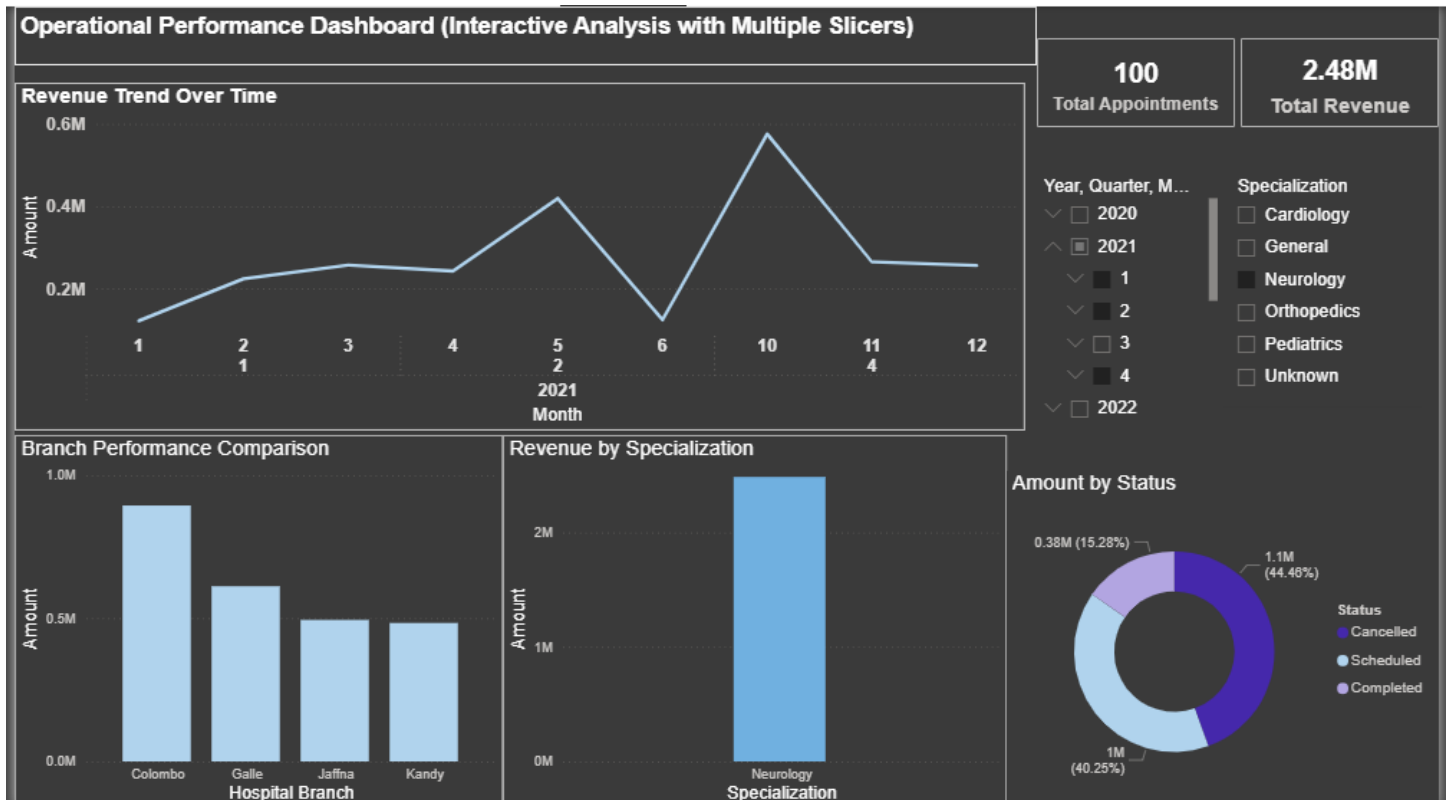
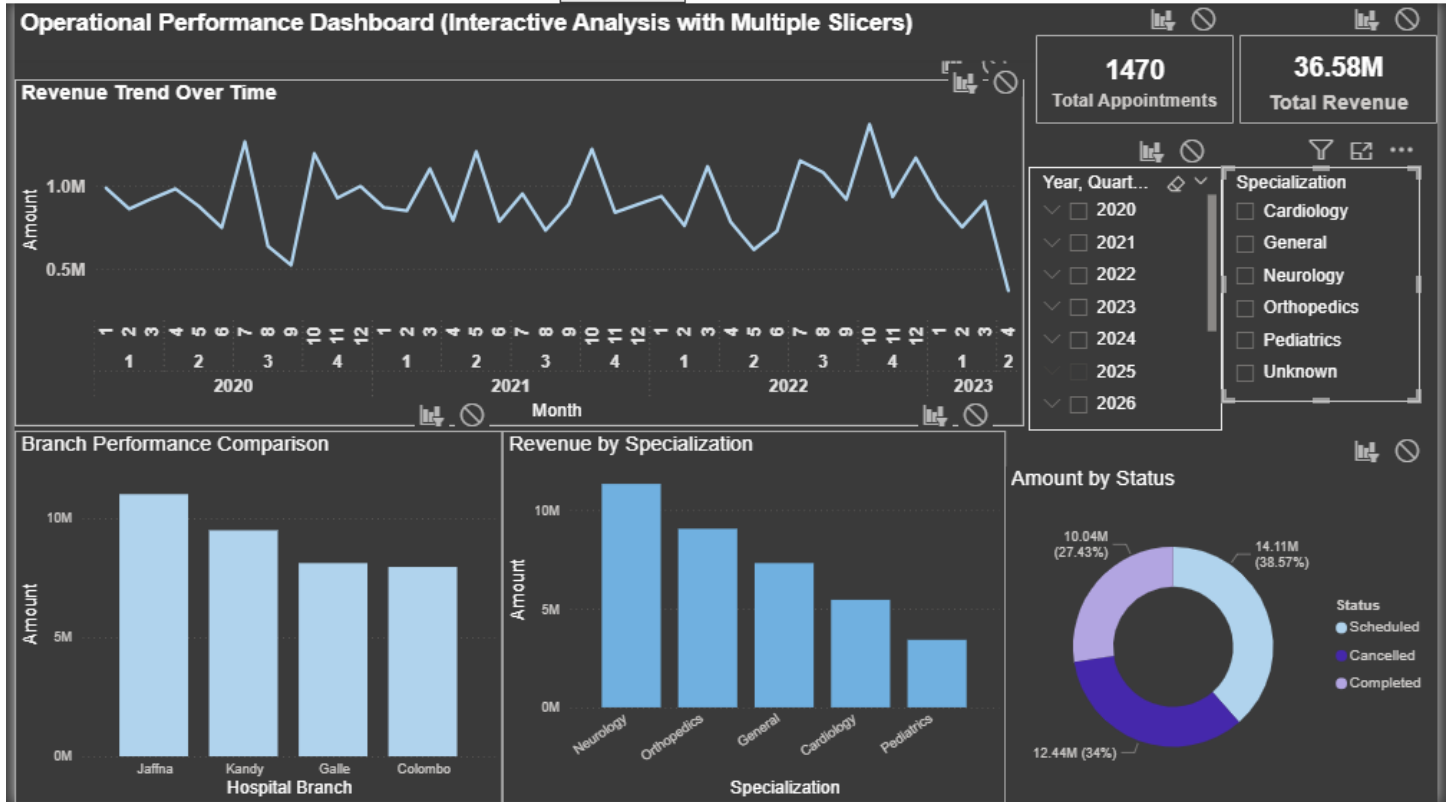
- Which hospital branch generates the highest revenue?
- Which specialization contributes most to revenue?
- How does revenue change over time?
- What is the distribution of appointment statuses?
- How do different filters affect performance?

Each visual in the dashboard serves a specific purpose and complements others:

1. Line Chart + Slicers → Shows how filtered data behaves over time
2. Branch Chart + Specialization Chart → Enables comparative analysis across dimensions
3. Donut Chart + KPI Cards → Provides proportional and summary insights

This demonstrates the use of multiple slicers, cascading filters, and interactive visuals to analyze hospital data.

By combining time-based trends, categorical comparisons, and proportional analysis, the dashboard provides a comprehensive view of operational performance. The use of interactivity enhances user experience and allows deeper insights into revenue patterns, patient behavior, and overall system efficiency.

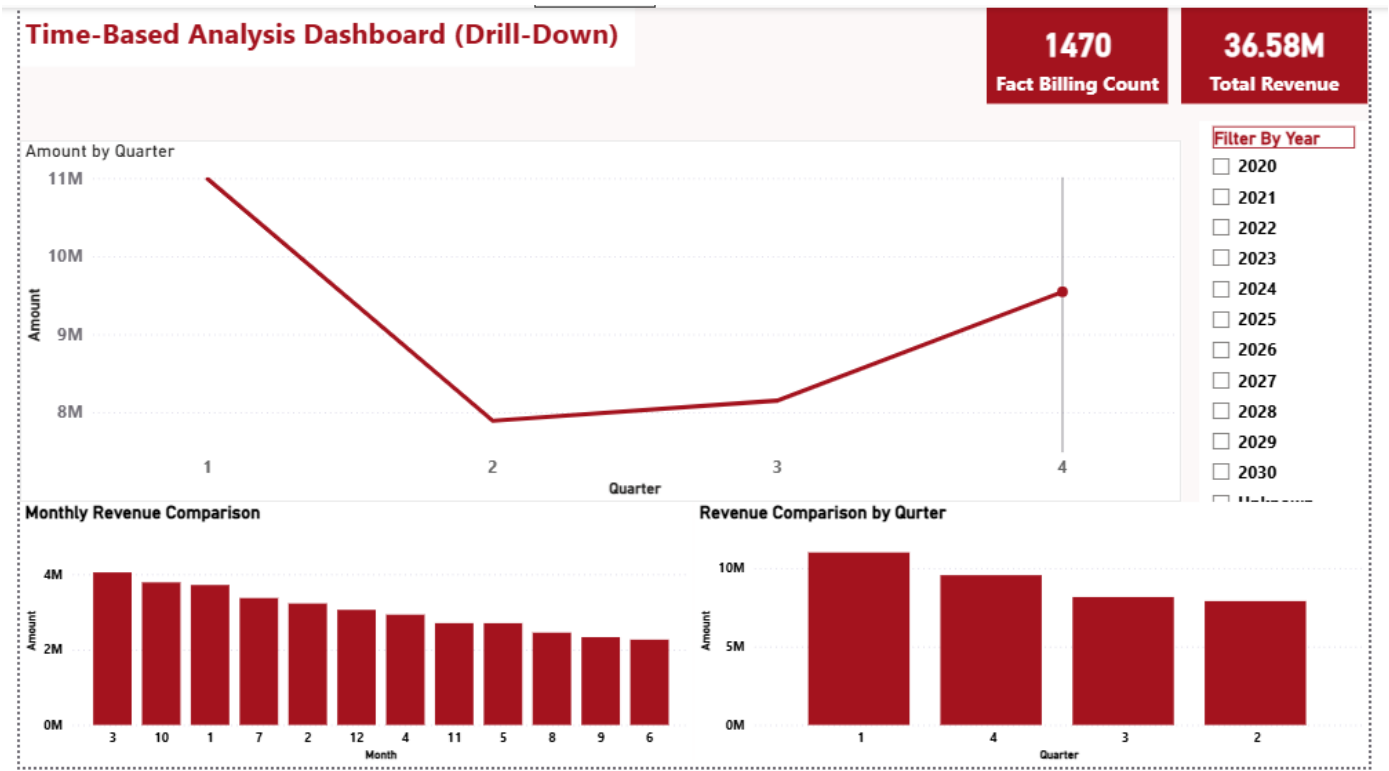


4.3 Report 3 - Time-Based Analysis Dashboard (Drill-Down)

This report focuses on analyzing hospital performance over time using a drill-down approach. It allows users to explore revenue and appointment behavior across different time levels, enabling a detailed understanding of when activity occurs.

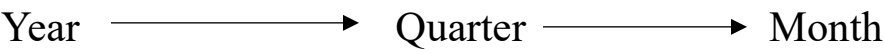
Dashboard Overview

The dashboard was designed to present time-based insights using hierarchical data. It combines a main drill-down visual with supporting charts and KPI indicators to provide both detailed and summarized analysis.



Drill-Down Implementation (Core Feature)

A hierarchy was created using the Date dimension:



This hierarchy enables users to navigate from high-level summaries to more detailed levels interactively.

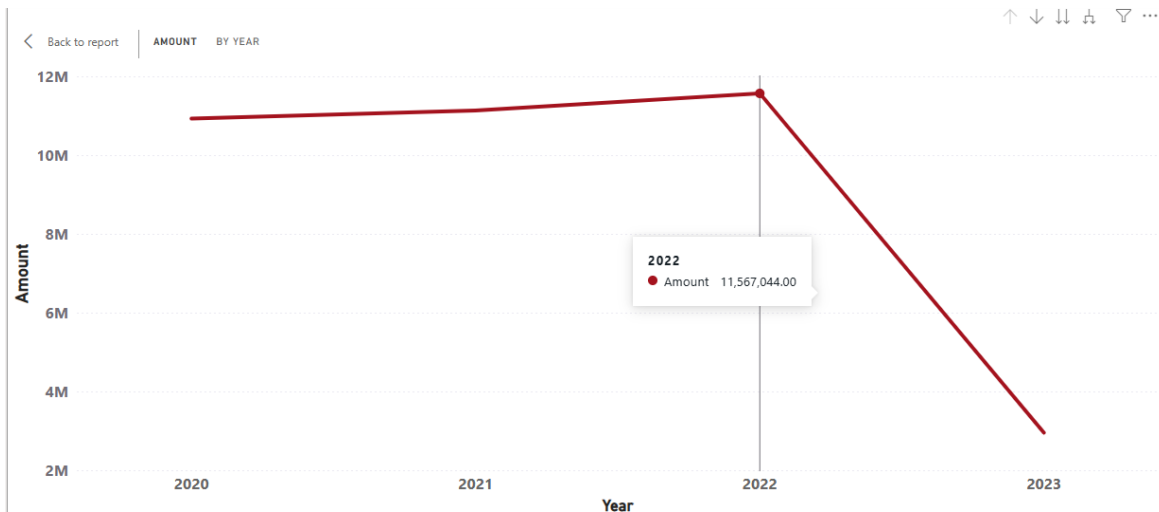
Line charts are ideal for time-based data as they clearly show trends, patterns, and fluctuations.

- Identifies revenue growth and decline over time
- Highlights peak periods and seasonal variations

1 – Year-Level Drill-Down View

The first view displays revenue aggregated at the year level . Users can observe overall performance trends across multiple years.

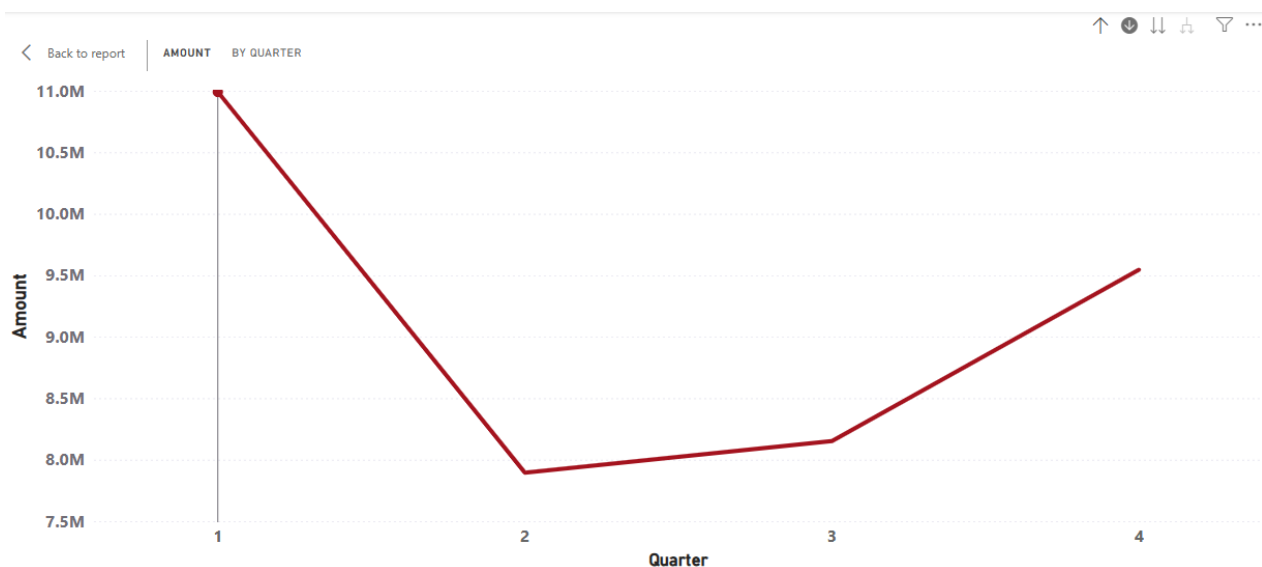
- Helps identify which year generated the highest revenue
- Provides a high-level comparison of performance



2 – Quater-Level Drill-Down View

After selecting a year, the report drills down to the quarter level.

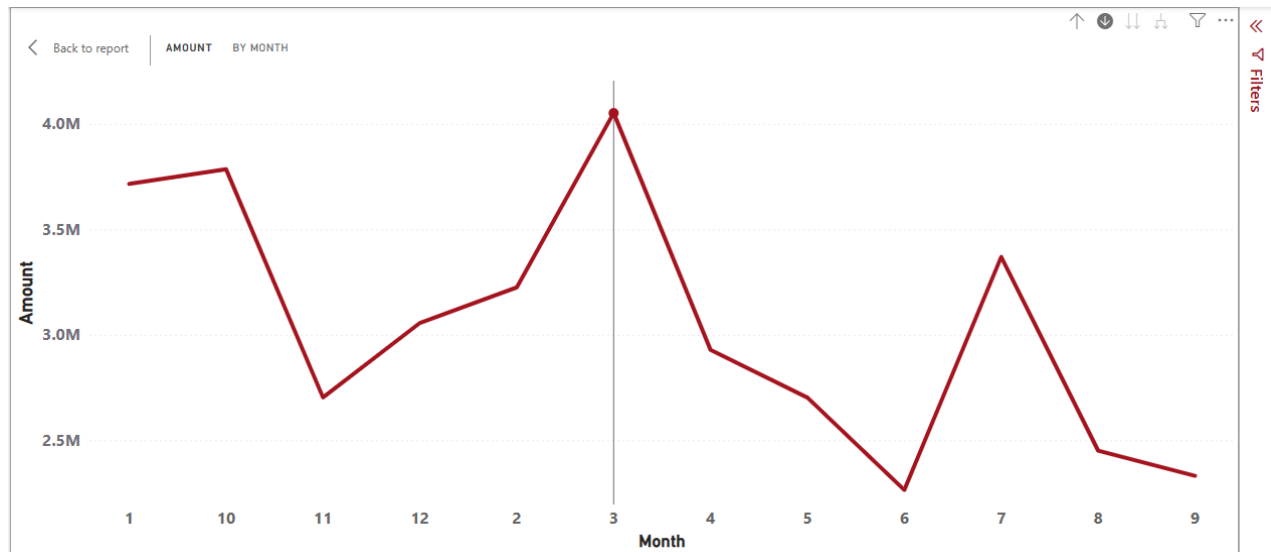
- Shows how revenue is distributed across quarters
- Helps identify strong and weak quarters within a year



3 –Month-Level Drill-Down View

Further drilling down displays data at the monthly level.

- Identifies peak months and low-performing months
- Reveals detailed patterns and fluctuations



Conclusion of Drill-Down:

This step-by-step navigation allows users to move from general trends to detailed analysis, making it easier to understand time-based behavior.

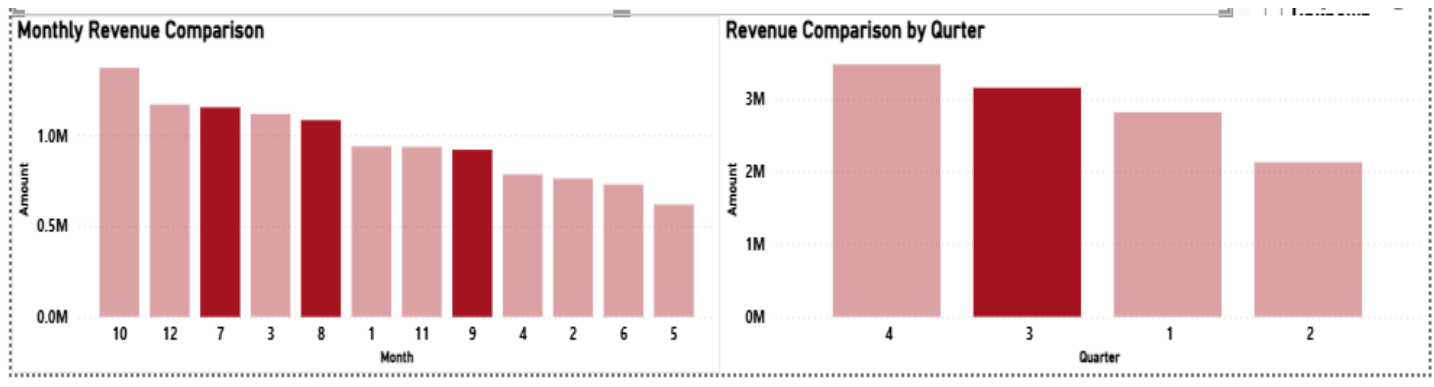
Column Chart – Monthly Revenue Comparison + Revenue Comparison by Quarters

A column chart was used to compare revenue across months or Quarter.

- Axis → Month or Quarter
- Values → Revenue (Amount)

Column charts are effective for comparing values across categories (months/Quarter)

- Helps identify highest and lowest revenue months
- Supports detailed comparison within a year
- Highlights which quarters perform better
- Helps identify seasonal trends



Interaction Between Visuals

All visuals are interconnected. When a user selects a specific year or interacts with a chart:

- Line chart updates dynamically
- Column charts adjust accordingly
- KPI values are recalculated

This creates a fully interactive analytical environment.

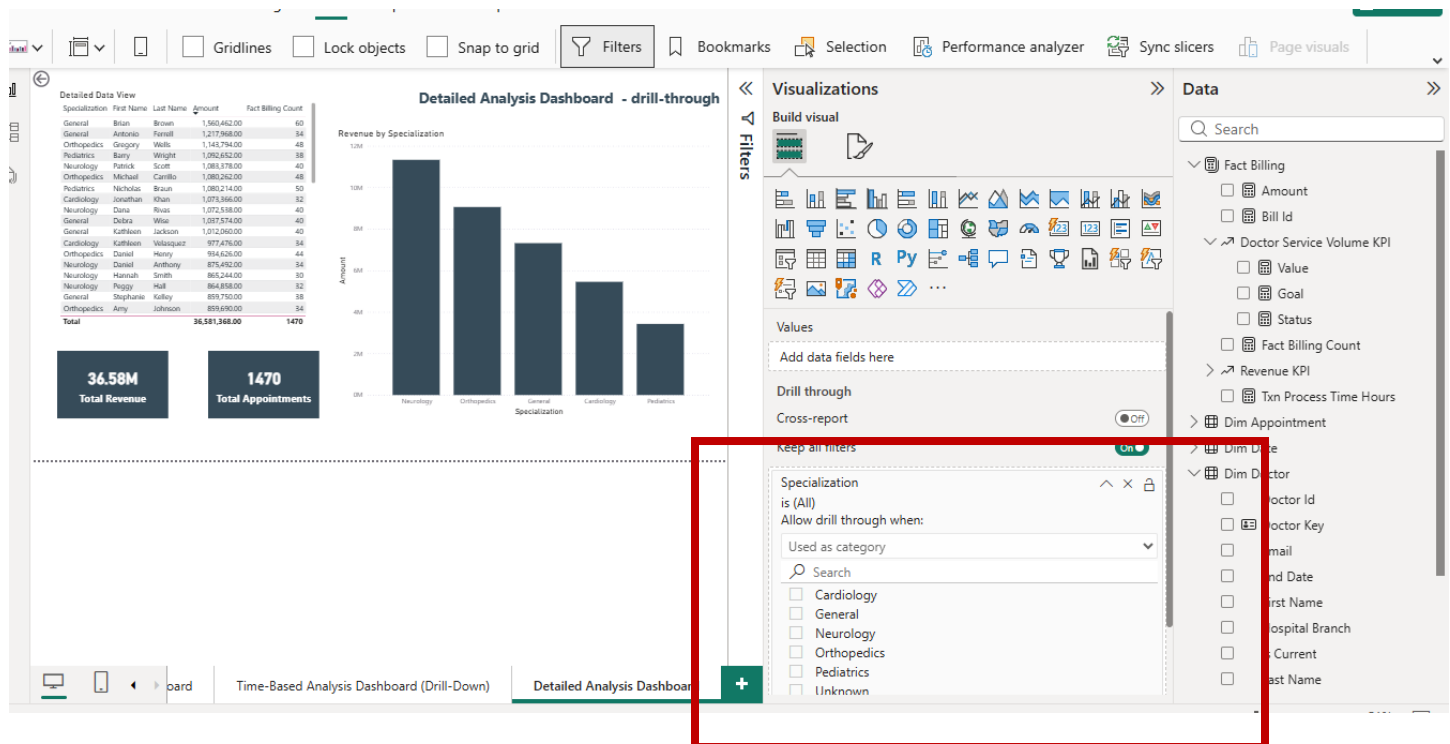
The drill-down dashboard successfully demonstrates hierarchical exploration of data using the Date dimension. By combining line charts, column charts, KPI indicators, and slicers, the report provides both high-level trends and detailed insights into time-based performance.

This approach enhances data understanding and supports better decision-making by revealing patterns, trends, and peak activity periods within the hospital system.

4.4 Report 4 - Detailed Analysis Dashboard (Drill-Through)

This report was developed to provide a detailed view of hospital performance using the drill-through feature in Power BI. It enables users to navigate from summary-level visuals to a dedicated page containing detailed and filtered information related to a selected data point.

Step 1 – Configuring Drill-Through Field



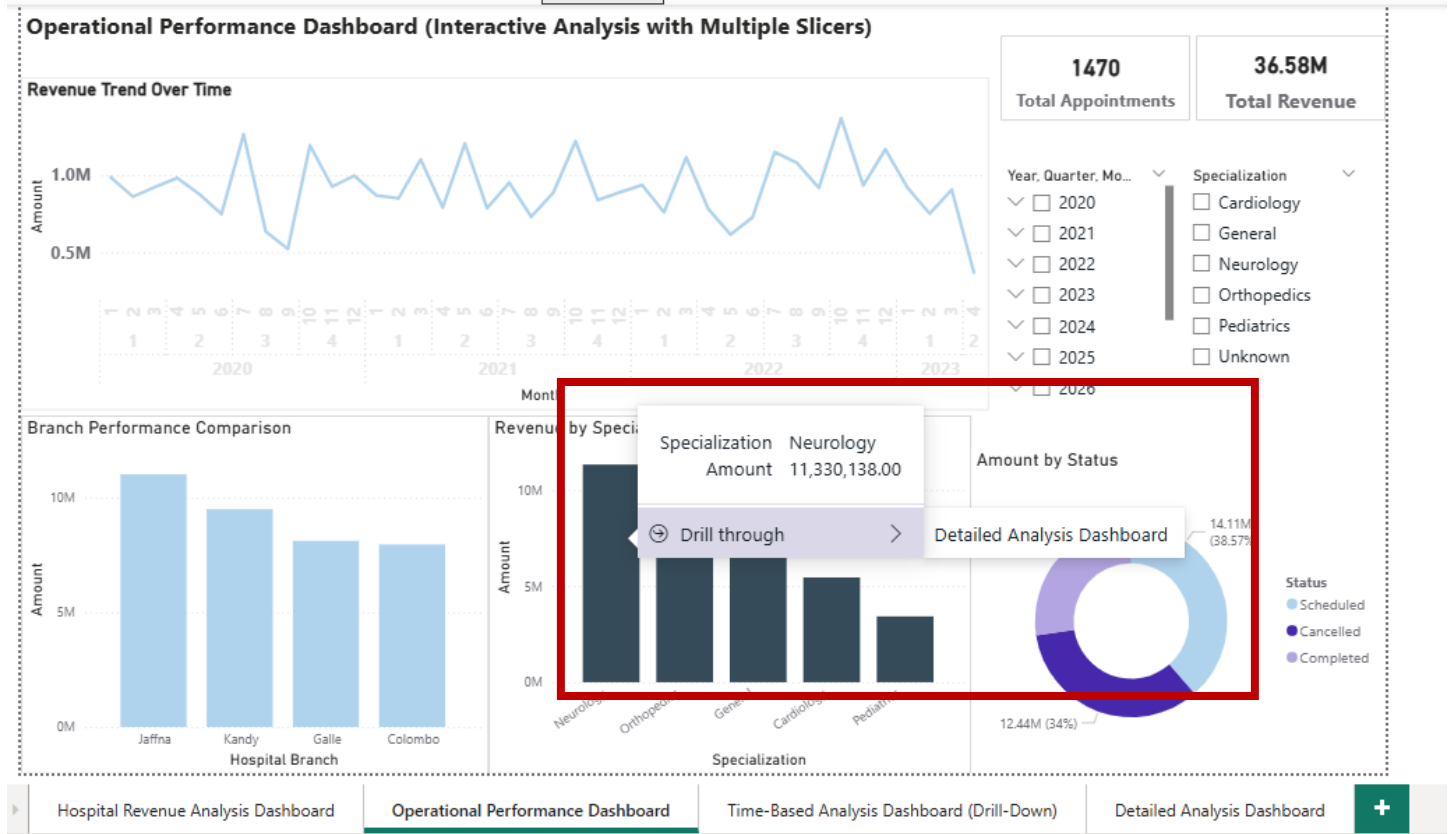
In this step, the Doctor Specialization field was added to the drill-through section in the Filters panel.

This configuration enables the report to receive filter context from other visuals. When a user selects a specialization in another report, this page will automatically display data related to that selection.

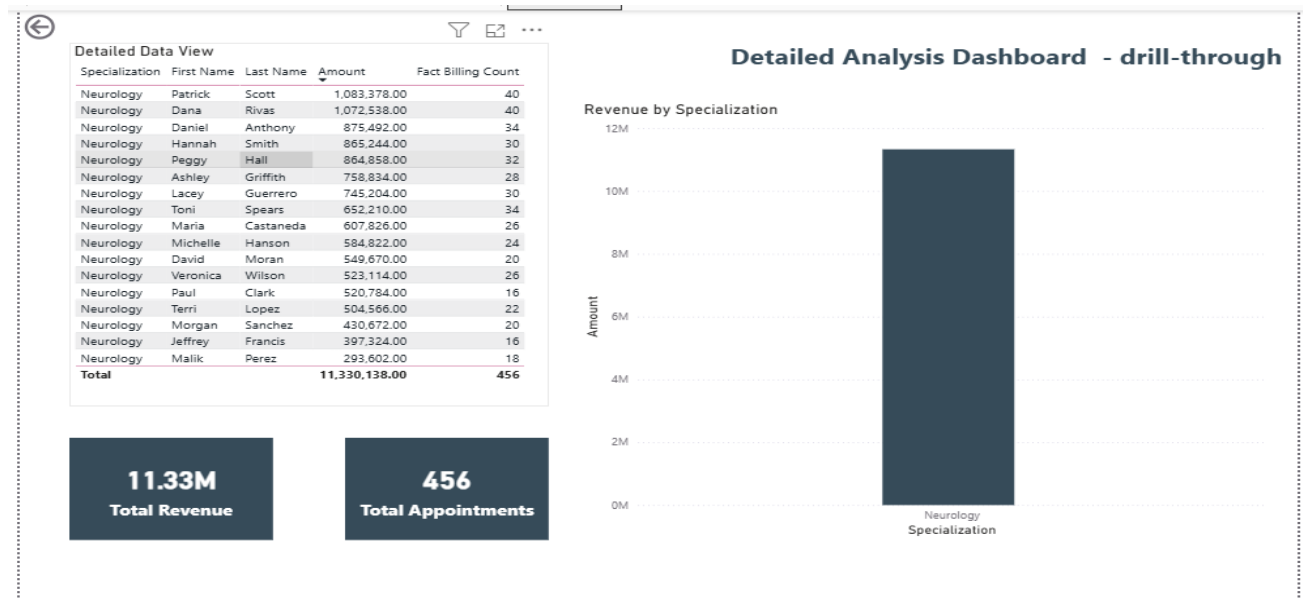
Step 2 – Initiating Drill-Through from Main Dashboard

From the Operational Performance Dashboard, a bar representing a specific specialization (e.g., Neurology) was selected. By right-clicking on the data point, the drill-through option was accessed and the user navigated to the detailed report page.

To allow users to move from a summarized visual to a detailed analysis page using contextual interaction.



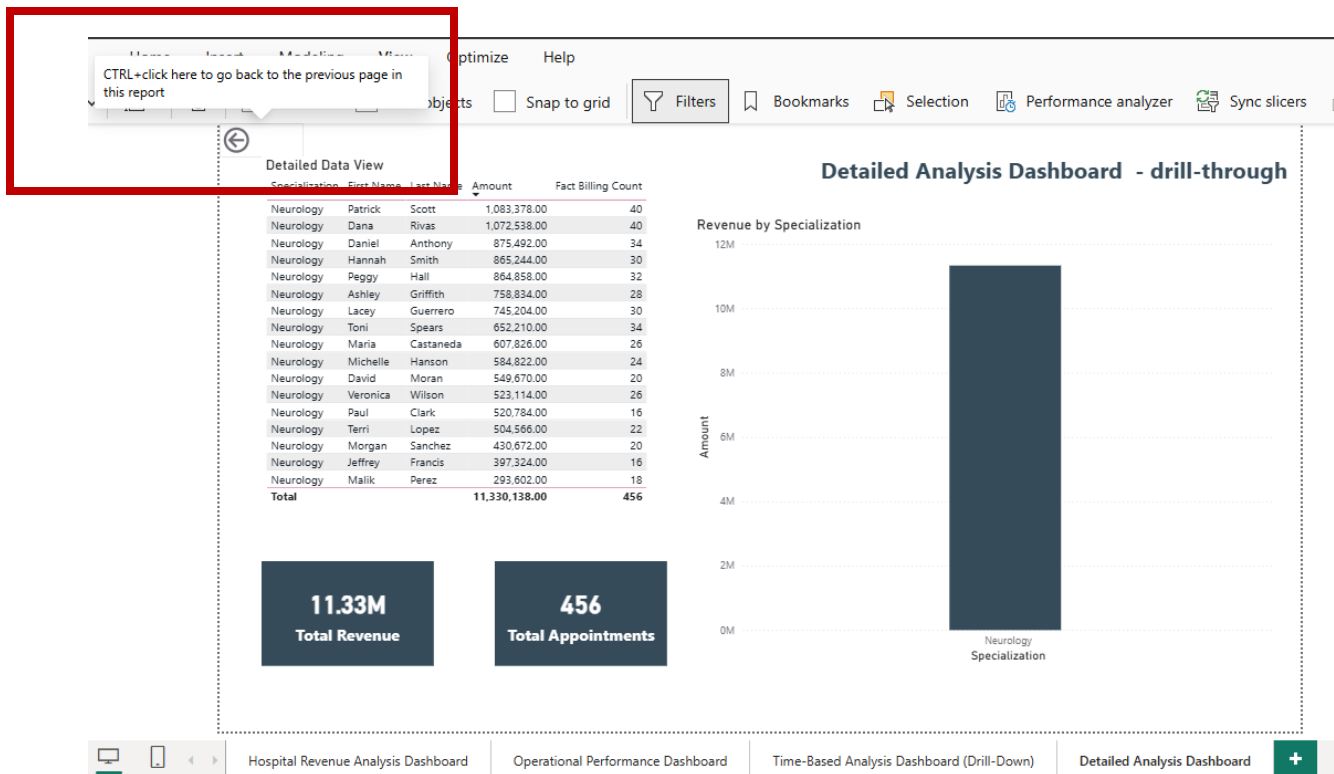
Step 3 – Drill-Through Navigation Result



After performing the drill-through action, the report navigates to the Detailed Analysis Dashboard, where all visuals are automatically filtered based on the selected specialization. For example, selecting “Neurology” displays only Neurology-related data across all visuals.

This is to provide a focused and filtered view of detailed data without manually applying filters.

Step 4 – Back Navigation



A back button was included at the top of the page. This button allows users to return to the previous report page after viewing detailed information.

The button works only when the page is accessed through drill-through navigation.

To improve usability and provide smooth navigation between summary and detailed reports.

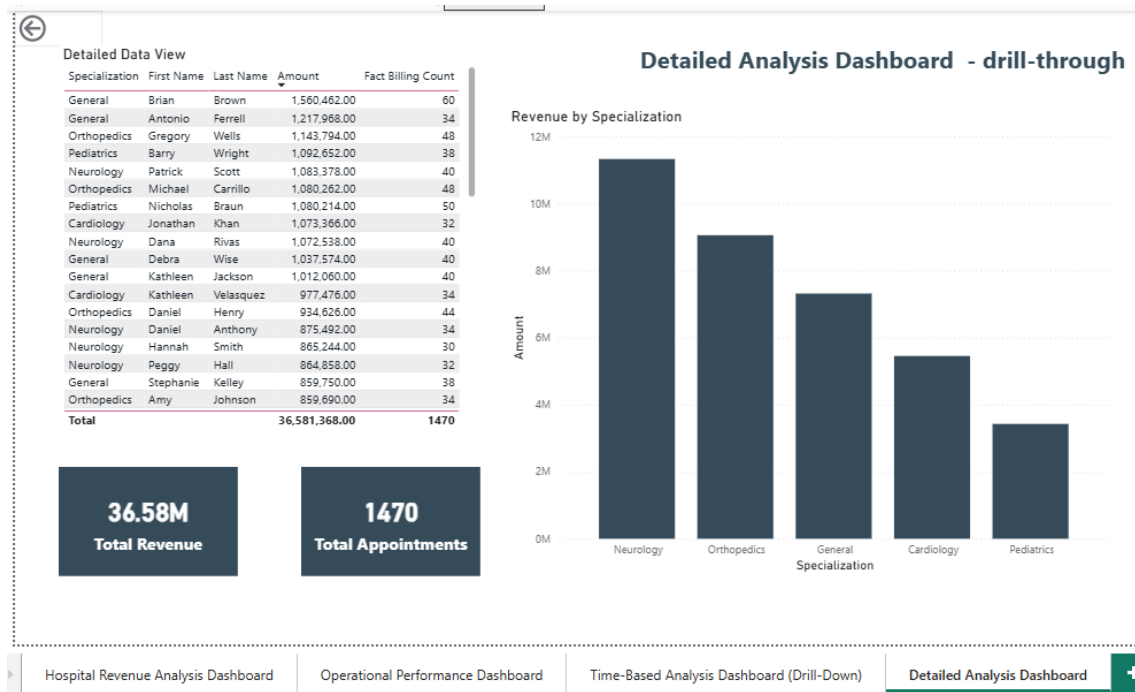
Step 5 – Visual Design (Detailed Data Presentation)

To present detailed insights, multiple visuals were combined within the drill-through page.

A table visual was used to display granular, row-level data including doctor specialization, doctor names, revenue (Amount), and appointment count. This allows users to examine individual records in detail.

In addition, a bar chart was created to represent revenue by specialization, where the specialization is placed on the axis and revenue (Amount) is used as the value. This helps to visually highlight the contribution of the selected specialization.

Furthermore, KPI cards were included to summarize key metrics such as total revenue and total appointments. These indicators provide a quick overview of performance for the selected context.



Data Preparation and Modeling

The report was built using data from the SSAS cube, which contains pre-defined relationships between fact and dimension tables. Measures such as revenue and appointment count were directly used from the cube.

Since the cube already provides calculated measures, no additional DAX expressions were required in this report.

Purpose

The purpose of this report is to allow users to explore detailed information about specific entities such as doctor specialization, improving analytical depth without cluttering the main dashboard.

Conclusion

The drill-through report successfully demonstrates advanced navigation and detailed data exploration in Power BI. By integrating interactive navigation, detailed tables, and visual summaries, the report enhances usability and supports effective decision-making.

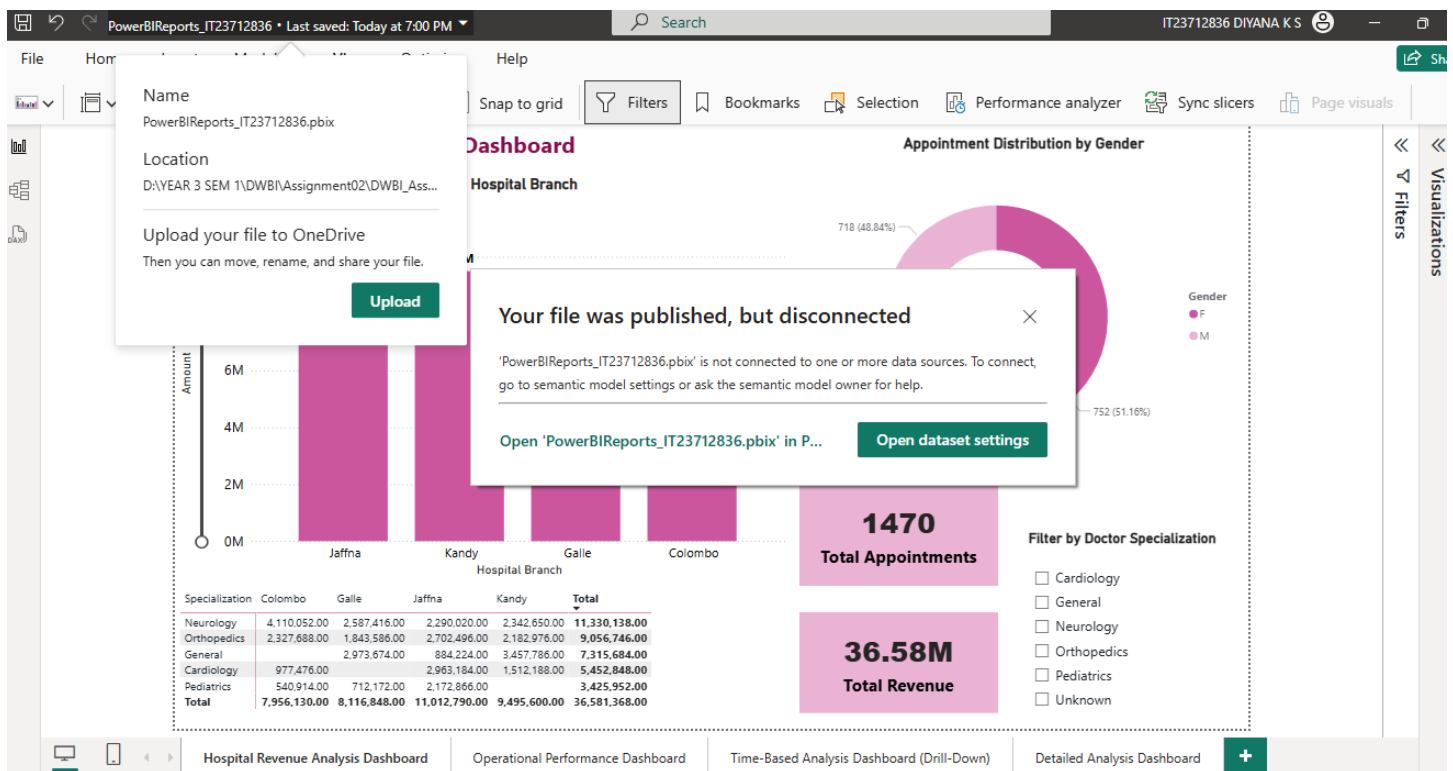
Report Publishing in Power BI Service

The developed Power BI reports were successfully published to Power BI Service using the publish feature available in Power BI Desktop.

After publishing, the reports were accessed through the Power BI Service web interface (online platform). All report pages, including matrix, interactive slicer-based, drill-down, and drill-through reports, were available in the workspace.

However, since the data source used in this project is an SSAS cube hosted locally, the Power BI Service could not establish a live connection to the data source without a configured gateway. As a result, the report appears as “*published but disconnected*” in the online environment.

Despite this limitation, the report structure, visuals, and navigation features were successfully deployed and demonstrated in Power BI Service.



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Detailed Analysis Dashboard

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Hospital Revenue Analysis Dashboard